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Predicting Financial Bankruptcy Using Machine Learning

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Table of Contents

1	Introduction.....	1
2	Literature Review.....	3
3	Model	6
3.1	Feature Selection.....	6
3.1.1	Correlation Matrix.....	6
3.1.2	ANOVA F-tests.....	7
3.2	Bankruptcy Prediction Model	8
3.2.1	Random Forest.....	9
3.2.2	XGBoost	10
3.2.3	LightGBM.....	11
3.3	Explainable AI	12
3.3.1	SHAP	12
4	Dataset and Statistical Analysis.....	12
4.1	Data Source and Dataset	13
4.2	Data Processing.....	13
4.3	Exploratory Data Analysis	15
5	Methodology	18
5.1	Random Forest	18
5.1.1	Hyperparameter Tuning	18
5.1.2	Evaluation Matrices	19
5.1.3	K-fold Cross Validation	19
5.1.4	Model Explain with SHAP	20
5.2	XGBoost	22
5.2.1	Hyperparameter Tuning	22
5.2.2	Evaluation Matrices	23
5.2.3	K-fold Cross Validation	23
5.2.4	Model Explain with SHAP	24
5.3	LightGBM.....	26
5.3.1	Hyperparameter Tuning	26
5.3.2	Evaluation Matrices	26
5.3.3	K-fold Cross Validation	27
5.3.4	Model Explain with SHAP	27

6	Thesis Discussion.....	29
6.1	Feature Selection.....	30
6.2	Bankruptcy Prediction Model	31
6.3	Explainable AI	33
7	Practical Implication	35
8	Conclusion	36
	References.....	38
	Appendix.....	43
	Declaration of Authenticity.....	63

ABSTRACT

Financial bankruptcy prediction has become a critical focus in risk management and corporate finance, driving the need for robust, data-driven decision-making frameworks. This thesis presents a machine learning-based approach that integrates feature selection, predictive modeling, and explainable AI (XAI) to enhance bankruptcy forecasting. The study explores two distinct modeling strategies: manual feature selection using statistical techniques and SHAP-driven feature importance, allowing a comparison between traditional selection methods and AI-driven explainability. By implementing Random Forest, XGBoost, and LightGBM, this study evaluates how well machine learning models predict financial distress. Additionally, the study emphasizes model interpretability, leveraging SHAP to explain the key financial indicators influencing bankruptcy predictions. This framework not only contributes to improved bankruptcy risk assessment but also provides actionable insights for financial analysts, businesses, and policymakers. The findings offer a scalable, interpretable, and adaptable approach for integrating machine learning in financial decision-making.

1 INTRODUCTION

Financial bankruptcy prediction is a crucial area of study for businesses, investors, and policymakers, as it helps in mitigating economic risks and improving financial stability. Traditional prediction methods such as logistical regression and discriminant modelling are based on financial ratios, but they often fail to capture the complex, nonlinear relationships seen in financial data. Machine learning (ML) offers a powerful alternative, leveraging computational techniques to enhance predictive accuracy and uncover patterns in large datasets. However, despite the potential of ML models like XGBoost, LightGBM, and Random Forest, several challenges persist, including class imbalance, feature selection, and model interpretability.

The class imbalance, in which bankrupt enterprises are substantially less numerous than non-bankrupt ones, is a serious problem in bankruptcy datasets and causes biased predictions. Additionally, selecting relevant financial features is critical to improving model performance, as redundant or irrelevant variables can lead to overfitting and increased computational costs. Another significant drawback of machine learning model is their inability to be interpreted; they frequently operate like "black boxes," thereby rendering it difficult for financial

professionals to understand their forecasts. Transparency in decision-making of finance is essential for regulatory compliance and stakeholder trust, underscoring the need of explainable AI (XAI) methodologies.

To address these challenges, this study develops a machine learning framework integrating feature selection, ensemble learning, and explainable AI techniques. Two modelling strategies are explored: one employing feature selection techniques such as ANOVA F-tests and correlation matrices before training the models, and another using Shapley Additive Explanations (SHAP) to dynamically determine feature importance. This allows for a comparative analysis of traditional statistical feature selection versus AI-driven feature importance ranking.

The research implements three ML models - Random Forest, XGBoost, and LightGBM - using a dataset from the Taiwan Economic Journal (1999–2009), which includes financial ratios. The Synthetic Minority Over-Sampling Technique (SMOTE) is used to rectify the dataset's imbalance. Metrics like the Classification Report, Confusion Matrix, ROC-AUC, and Precision-Recall AUC are used to assess how well a model performs. To provide generalizability and robustness, K-fold cross-validation is implemented.

Beyond prediction accuracy, the study places a strong emphasis on model interpretability by leveraging SHAP to explain the key financial indicators influencing bankruptcy risk. This enhances model transparency, making the results more actionable for financial analysts, regulators, and decision-makers. By integrating feature selection, predictive modelling, and explainable AI, this research offers a scalable and interpretable approach to bankruptcy risk assessment.

This thesis is organized as follows:

Chapter 2 reviews traditional and modern bankruptcy prediction models, emphasizing explainable AI in financial risk assessment.

Chapter 3 details the model's introduction including feature selection techniques, bankruptcy prediction models, and SHAP.

Chapter 4 discusses the dataset, data processing, and exploratory data analysis.

Chapter 5 explains methodological framework of ML models including hyperparameter tuning, evaluation metrics, K-fold cross-validation, and model explanation with SHAP.

Chapter 6 evaluates the thesis and connects with traditional approaches based on feature selection, prediction model, and Explainable AI.

Chapter 7 highlights the issues and discusses practical implementation of the research.

Chapter 8 concludes with key insights.

This study improves bankruptcy prediction models by combining explainable AI, machine learning, and feature selection, thereby reducing the disparity between interpretability and forecast accuracy. The findings offer valuable applications in financial risk assessment, corporate governance, and investment decision-making.

2 LITERATURE REVIEW

The estimation of financial bankruptcy has been a critical area of research in finance and machine learning, as the early identification of risky firms can help mitigate economic losses and improve financial stability. Over the years, various methodologies, ranging from statistical models to advanced machine learning algorithms, have been explored to improve bankruptcy prediction accuracy. The main scientific contributions are summarized in this review of the literature, together with their methodology, feature selection strategies, and applications of explainable AI (XAI).

Feature Selection in Bankruptcy Prediction

Feature selection is crucial in bankruptcy prediction models, as selecting relevant financial indicators can significantly impact model performance. **Barboza et al. (2017)** investigated financial ratios as features for predicting bankruptcy and applied machine learning models like Random Forest, Support Vector Machines (SVM), and Neural Networks. Similarly, **Shetty et al. (2022)** implemented statistical feature selection techniques and applied Gradient Boosting and XGBoost models.

Qu et al. (2019) focused on feature importance analysis and used Deep Learning and SVM for bankruptcy prediction. Meanwhile, **Lin et al. (2011)** employed Principal Component Analysis (PCA) to reduce dimensionality before training Neural Networks and SVM models. Another approach was proposed by **Pompe & Feelders (1997)**, who used statistical analysis to identify significant financial ratios and applied Neural Networks and Decision Trees.

More recent research has explored hybrid selection techniques. **Antulov-Fantulin et al. (2021)** combined financial and governance indicators and applied Random Forest and Decision Trees. **Yu et al. (2013)** relied on expert-selected financial ratios for bankruptcy prediction using Extreme Learning Machines. By integrating financial news and reports into Deep Learning models - more especially, Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks, **Mai et al. (2018)** introduced textual data analysis.

Other studies have focused on systematic feature selection. **Dasilas & Rigani (2024)** applied systematic feature selection across multiple machine learning models, while **Petropoulos et al. (2020)** explored ensemble learning techniques with financial ratios. **Siswoyo et al. (2022)** proposed an ensemble feature selection technique, integrating multiple boosting algorithms to enhance bankruptcy prediction accuracy.

Machine Learning Models for Bankruptcy Prediction

From deep learning techniques to traditional methods, a variety of machine learning models have been used to forecast bankruptcy. **Le & Viviani (2017)** and **Zhang (2017)** employed logistic regression and SVM models using financial ratios. Similarly, **Bonello et al. (2018)** investigated hybrid feature selection and applied machine learning-stochastic hybrid models.

Deep learning techniques have also been explored extensively. **Park et al. (2021)** and **Kim et al. (2021)** used Recurrent Neural Networks (RNN) and SHAP-based explainability to understand their decision-making process. **Liang et al. (2016)** developed hybrid machine learning models that incorporated corporate governance indicators. **Smiti & Soui (2020)** applied Synthetic Minority Over-Sampling Technique (SMOTE) combined with a Stacked Autoencoder for feature extraction. In order to predict bankruptcy, **Vochozka et al. (2020)** suggested using Long Short-Term Memory (LSTM) networks to process sequential financial data.

Ensemble learning techniques have also proven effective. **Zhao et al. (2016)** introduced Kernel Extreme Learning Machines (KELM) to improve prediction robustness. **Huang & Yen (2019)** applied feature engineering alongside Random Forest and SVM classifiers. **Radwan et al. (2022)** experimented with Deep Extreme Learning Machines, while **Özparlak et al. (2022)** explored Artificial Neural Networks (ANN) for corporate bankruptcy forecasting.

Explainable AI (XAI) in Bankruptcy Prediction

Despite advancements in predictive accuracy, interpretability remains a significant challenge in financial bankruptcy prediction. To increase transparency, models have incorporated explainable AI approaches like SHAP (Shapley Additive Explanations). **Park et al. (2021)** and **Kim et al. (2021)** applied SHAP to interpret neural network decisions and provide insight into feature importance.

Although many studies focus on enhancing prediction accuracy, few emphasize explainability. **Iparraguirre-Villanueva & Cabanillas-Carbonell (2024)** developed a Stacking Classifier model but did not apply explainability techniques. **Chen et al. (2023)** explored text-based features using XGBoost and Random Forest but did not provide interpretability measures. **Shi & Li (2019)** directed a thorough analysis of existing bankruptcy prediction models but noted a lack of studies incorporating explainability.

The literature review emphasizes the variety of approaches—from sophisticated machine learning models to statistical techniques—that are employed in bankruptcy prediction. Feature selection methods play a crucial role in improving model efficiency, with studies employing statistical analysis, PCA, expert-selected features, and hybrid approaches. Machine learning models including Random Forest, XGBoost, Deep Learning, and ensemble learning techniques have been widely adopted, demonstrating high predictive accuracy. However, the need for interpretability remains a major challenge, with only a few studies incorporating Explainable AI techniques such as SHAP. In this research, the focus on integrating interpretability frameworks to enhance model trustworthiness aims to identify crucial features for financial bankruptcy prediction, ensuring transparency and improving the reliability of machine learning models in financial decision-making.

3 MODEL

This chapter introduces the framework for machine learning-based financial bankruptcy prediction. The three main elements of the chapter are Feature Selection, Bankruptcy Prediction Models, and Explainable AI (XAI).

3.1 FEATURE SELECTION

By finding the most pertinent financial indicators and removing superfluous or unnecessary information, feature selection is essential to enhancing the predictive performance of machine learning models. **Barboza et al. (2017)** investigated a number of machine learning methods for predicting bankruptcy and stressed how crucial it is to use relevant financial ratios in order to improve model performance. In this study, two distinct approaches to feature selection have been implemented: (1) Traditional feature selection techniques such as Correlation Matrix and ANOVA F-tests, and (2) SHAP-based feature selection, which allows the model to dynamically determine the most impactful features.

3.1.1 CORRELATION MATRIX

A statistical technique for assessing the linear link between financial data and a company's bankruptcy state is the correlation matrix. **Zhang (2017)** demonstrated that selecting financially significant ratios using correlation-based filtering improves the predictive power of machine learning models while reducing overfitting. In this study, the degree of linear relationship between each attribute and the target variable (bankruptcy) was calculated using Pearson's correlation coefficient. The correlation heatmap (Figure 1) visually represents these relationships, making it easier to detect patterns and dependencies among variables.

To refine our feature set, we established a correlation threshold of 0.2 (absolute value), ensuring that only features with a meaningful relationship to bankruptcy were retained. The selected features include: Per Share Net profit before tax, Net worth/Assets, Debt ratio, Retained Earnings to Total Assets, Net Income to Total Assets, Net profit before tax/Paid-in capital, and Persistent EPS in the Last Four Seasons.

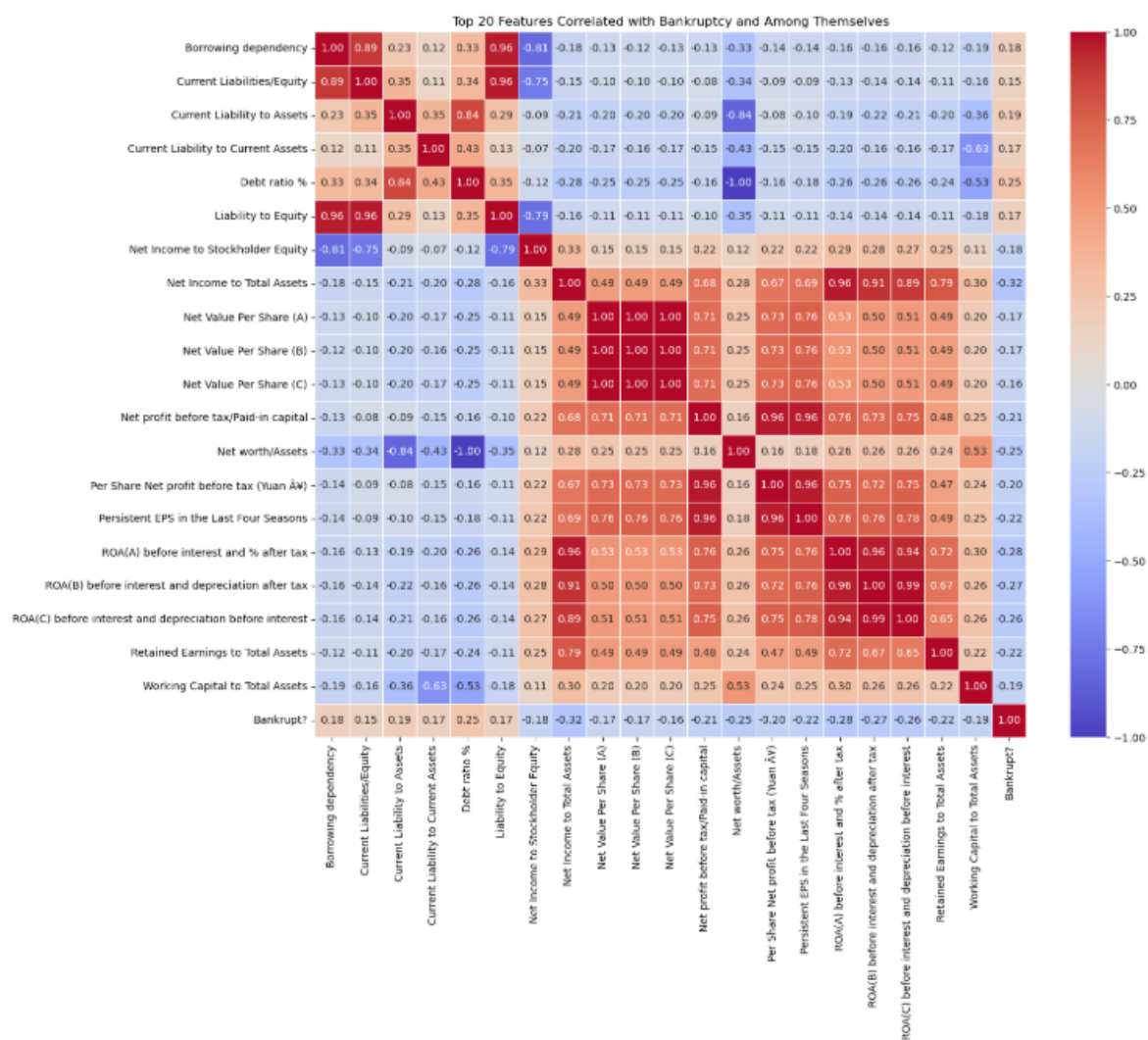


Figure 1: Top 20 Features Correlated with Bankruptcy and among themselves

3.1.2 ANOVA F-TESTS

The Analysis of Variance (ANOVA) F-test is a commonly used statistical technique to determine if the means of many groups differ significantly from one another. In the paradigm of feature selection, the ANOVA F-test measures how well each independent variable (financial ratio) can distinguish between the two categories of the target variable (bankrupt and non-bankrupt firms). The F-statistic, which shows the ratio of variation between groups (bankrupt vs. non-bankrupt) to variance within groups, is computed to accomplish this. A higher F-value (Figure 2) indicates a stronger discriminatory power of a feature, suggesting its importance in classification models. **Pompe & Feelders (1997)** successfully applied ANOVA F-tests to select significant financial ratios that enhance bankruptcy prediction accuracy.

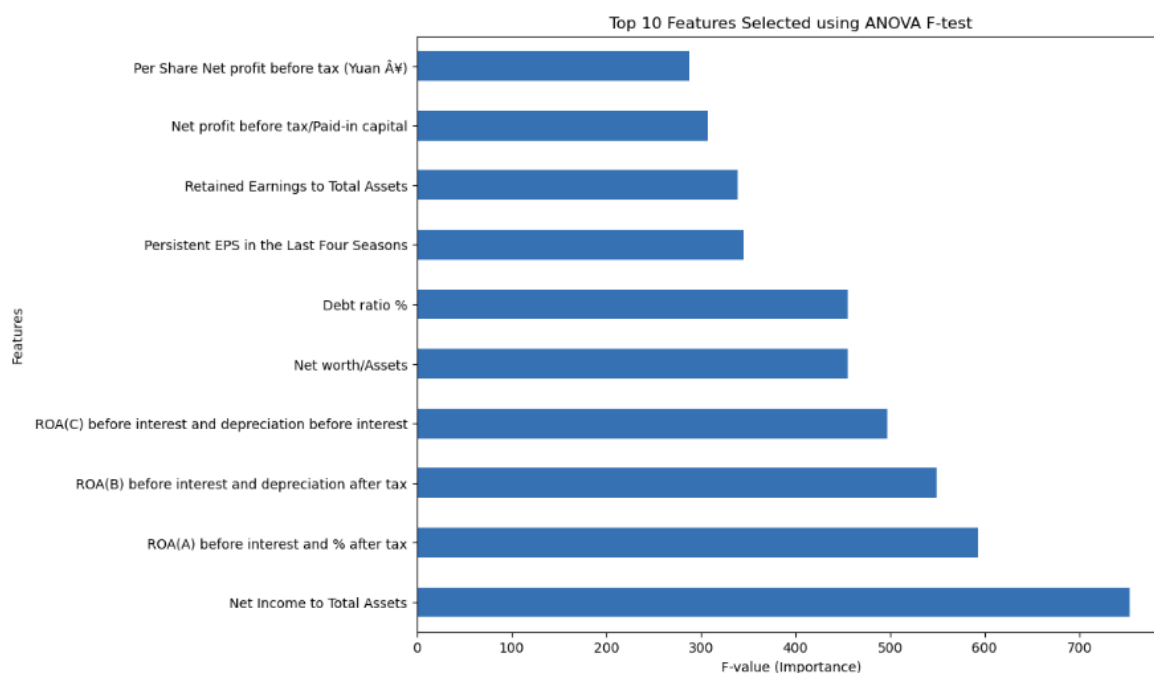


Figure 2: Top 10 Features Selected using ANOVA F-tests

However, financial data often exhibit complex, non-linear interactions that may not be adequately captured through this method (**Li & Wang, 2017**). Additionally, ANOVA F-tests treat features as independent variables, disregarding potential correlations among financial indicators. This is particularly problematic in bankruptcy prediction, where multiple profitability ratios and leverage measures are interrelated, leading to potential redundancy in feature selection (**Lombardo et al., 2022**).

3.2 BANKRUPTCY PREDICTION MODEL

Traditional statistical models, including logistical regression and discriminant modelling, have been used extensively to forecast bankruptcy, but they frequently have trouble analysing complicated, non-linear correlations in financial data (**Erdogan, 2012**). However, machine learning algorithms have become extremely effective tools that use massive datasets to find complex patterns and increase forecast accuracy. This research investigates three state-of-the-art machine learning models for bankruptcy prediction: Random Forest, XGBoost, and LightGBM.

3.2.1 RANDOM FOREST

The Random Forest (RF) ensemble learning method was created by **Breiman (2001)**. It creates numerous decision trees throughout training and integrates their forecasts in order to increase accuracy and reduce overfitting. Unlike a single decision tree, which might overfit on training data, Random Forest uses feature randomness and bagging (Bootstrap Aggregating) to produce different trees that together produce reliable predictions.

An example diagram showing how Random Forest builds several decision trees and aggregates their forecasts may be found below (Figure 3):

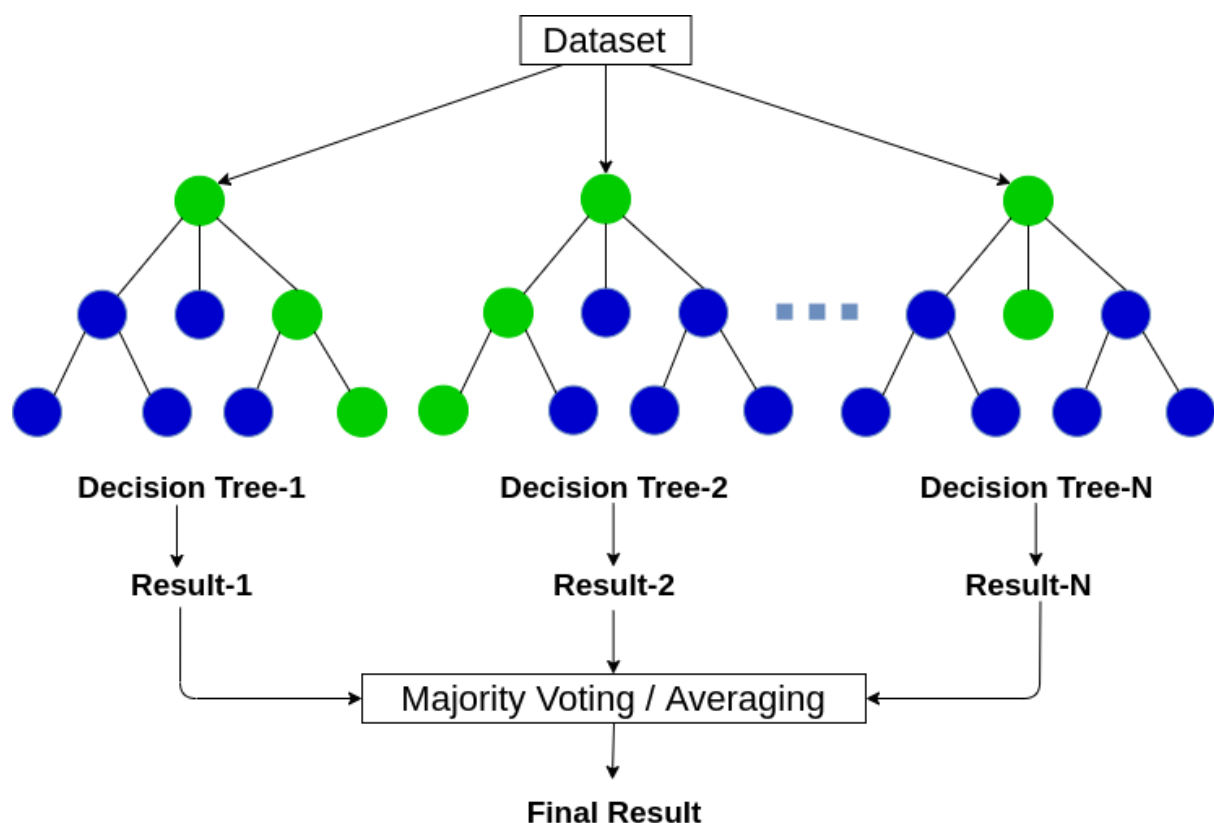


Figure 3: Illustrative diagram of Random Forest structure (source: datamahadev.com)

However, despite its strengths, Random Forest does have limitations. It may become computationally expensive with large datasets, and its interpretability is relatively lower compared to simple decision trees, making it necessary to integrate Explainable AI techniques such as SHAP to enhance transparency in predictions (**Park et al., 2021**).

3.2.2 XGBOOST

Extreme Gradient Boosting (XGBoost) is a powerful machine learning method that is well-known for its efficacy and precision in forecasting results in a variety of domains, including financial risk assessment and bankruptcy prediction. By combining sophisticated regularization techniques and effective handling of missing variables, this improved gradient boosting approach improves computing speed and predictive performance. Unlike traditional ensemble models, XGBoost employs an additive model where trees are sequentially constructed, each correcting the errors of its predecessors. This iterative refinement process enables XGBoost to capture complex patterns in financial data and improve prediction accuracy. Figure 4 illustrates the fundamental working mechanism of XGBoost, showcasing how decision trees are built sequentially while minimizing residual errors at each step.

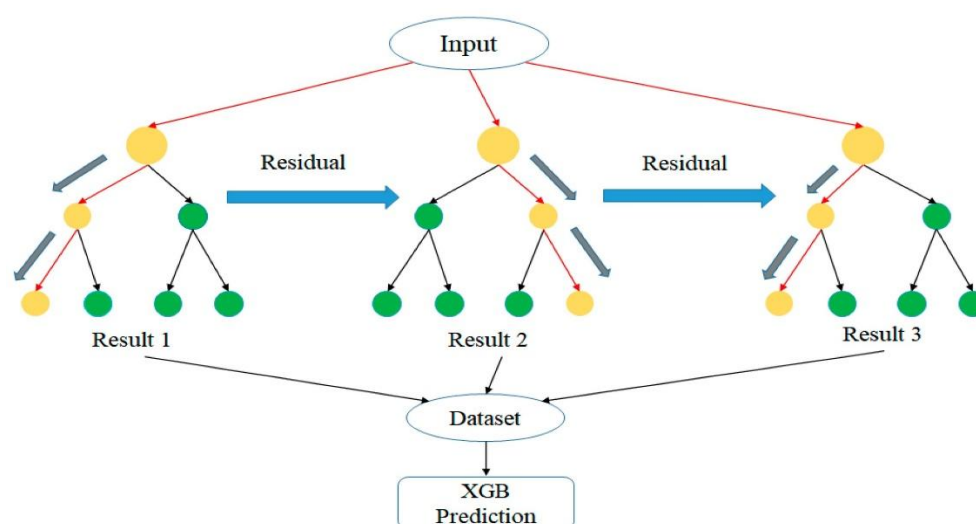


Figure 4: Illustrative diagram of XGBoost structure (source: mdpi.com)

XGBoost's ability to iteratively refine predictions and reduce errors makes it highly effective for complex problems like financial bankruptcy prediction. Studies such as **Radwan et al. (2022)** have demonstrated that since XGBoost can successfully handle feature interactions and imbalanced datasets, it performs better than traditional models like logistic regression and decision trees for predicting financial distress. Additionally, **Li & Wang (2017)** highlight that XGBoost's feature importance ranking provides valuable insights into key financial indicators contributing to bankruptcy, improving transparency and interpretability in financial risk assessment.

3.2.3 LIGHTGBM

Another gradient boosting framework created by Microsoft is Light Gradient Boosting Machine (LightGBM), which is intended to be extremely accurate, scalable, and efficient for classification and regression tasks. LightGBM employs a leaf-wise growth method, which optimizes decision trees by growing the leaf that delivers the biggest decrease in loss, resulting in deeper and more efficient trees, in contrast to typical tree-based models that build trees level by level (Figure 5).

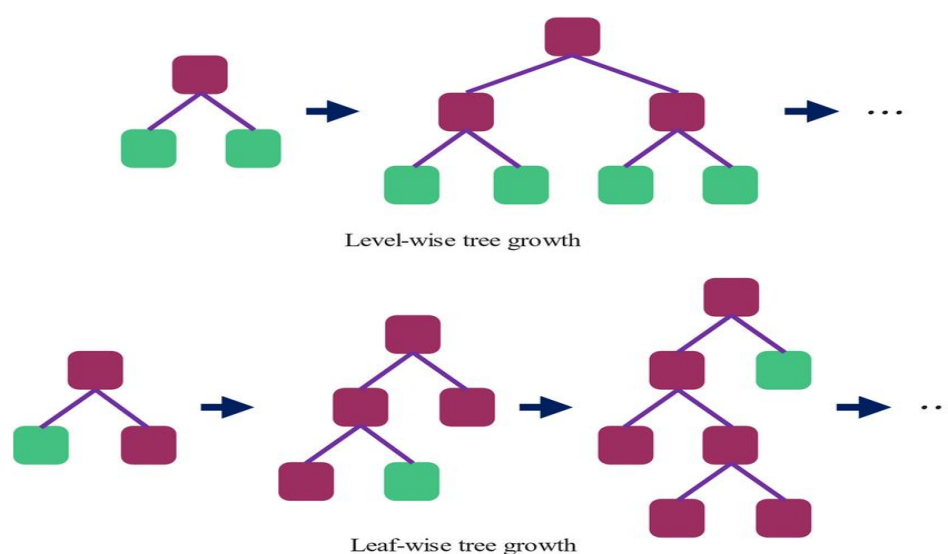


Figure 5: Level-wise vs. Leaf-wise Growth in Decision Trees (source: Ke et al., 2017)

Ke et al. (2017) introduced LightGBM as an enhanced variant of conventional gradient boosting, integrating methods including leaf-wise tree growth, histogram-based learning, and Gradient-Based One-Side Sampling (GOSS) to improve its predictive power. Unlike conventional boosting algorithms that evaluate all data points to determine splits, LightGBM optimizes tree-building by selecting only the most significant gradients, making it significantly faster in handling high-dimensional bankruptcy datasets. **Liu et al. (2021)** demonstrated that LightGBM's leaf-wise strategy improves classification performance in highly imbalanced financial datasets, as it enables the model to prioritize informative features in bankruptcy prediction.

3.3 EXPLAINABLE AI

As "black boxes," machine learning models—particularly ensemble techniques like Random Forest, XGBoost, and LightGBM—make precise predictions but do not always provide justifications for their choices. This lack of transparency limits their adoption in financial decision-making, where regulators require interpretable models to ensure fairness and accountability (**Lipton, 2018**). XAI approaches have been created to shed light on the significance of specific aspects in model predictions, especially post-hoc explanation methods. Among these techniques, Shapley Additive Explanations (SHAP) has emerged as one of the most effective frameworks for understanding feature contributions in complex machine learning models.

3.3.1 SHAP

Shapley Additive Explanations (SHAP) is a game-theoretic approach that can be used to explain the output of any machine learning model. It offers a consistent and equitable method of allocating each input feature's contribution to the model's calculations. Introduced by **Lundberg and Lee (2017)**, Cooperative game theory's Shapley values, which have been widely applied in economics to equitably distribute payouts among coalition members, are the foundation of SHAP. It ensures that the contribution of each feature is quantified across all potential feature combinations in machine learning through the assignment of worth to each feature in a given prediction.

In this thesis, SHAP is implemented in two distinct approaches: first, as an interpretability tool for evaluating models trained on statistically pre-selected features, and second, as a feature selection method that dynamically determines the most important financial indicators during model training.

4 DATASET AND STATISTICAL ANALYSIS

The dataset used to predict bankruptcy is thoroughly examined in this chapter, which also covers data sources, data pre-processing, and exploratory data analysis (EDA). The objective is to ensure data quality, remove inconsistencies, and extract meaningful insights before applying machine learning models.

4.1 DATA SOURCE AND DATASET

The Taiwan Economic Journal (TEJ) provided the dataset used in this study, which includes financial and operational information about Taiwanese publicly traded enterprises from 1999 to 2009. This dataset has been widely used in bankruptcy prediction research due to its comprehensive financial indicators and its applicability in real-world risk assessment. The Taiwan Stock Exchange's business requirements are followed when classifying corporate bankruptcy, guaranteeing that the ground truth labels for bankrupt and non-bankrupt enterprises correspond with official financial reporting standards.

The dataset consists of 95 independent financial variables along with the target variable, which indicates whether a company went bankrupt (denoted as 1) or remained solvent (denoted as 0). These financial metrics include cash flow indicators, operational efficiency metrics, profitability ratios, liquidity ratios, leverage ratios, and other measures of financial stability. Every component offers information about a company's financial situation and possible bankruptcy risk.

The inherent class imbalance in this dataset is a major problem because there are far fewer bankrupt enterprises than non-bankrupt firms. Corporate bankruptcy is a relatively uncommon occurrence in the real world, which is reflected in this. However, it also creates challenges for model training because predictive algorithms may become biased in favour of the majority class. This issue was later resolved by balancing the dataset and ensuring robust model performance using the Synthetic Minority Over-Sampling Technique (SMOTE).

4.2 DATA PROCESSING

The stage of data processing is essential to promising the accuracy and interpretability of the bankruptcy prediction models. This process involved handling missing values, selecting the most relevant financial features, and addressing class imbalance to improve model robustness and generalization.

A thorough assessment of missing values was conducted to determine whether any data imputation techniques were necessary. The dataset was found to be complete, with no missing financial indicators, eliminating the need for imputation. This aligns with findings by **Li and Wang (2017)**, who emphasized that datasets without missing values lead to improved model

performance in financial risk assessments. The absence of missing data allowed the focus to shift towards refining the dataset through feature selection and class balancing.

Feature selection was performed using a combination of correlation analysis and ANOVA F-tests to enhance computational efficiency. The correlation matrix heatmap was used to identify correlation of the target variable with financial ratios, revealing that most of the financial ratios show weak relationship with the target variable (Figure 01). Features with correlation coefficients exceeding a predefined threshold (0.2) were considered for further exploration. Additionally, ANOVA F-tests were applied to statistically assess which financial ratios showed the most significant disparities between bankrupt and non-bankrupt enterprises. Retaining only the most statistically significant features aligned with the approach recommended by **Erdogan (2012)**, who found that the discriminatory efficiency of machine learning models in predicting financial distress is enhanced by ANOVA-driven selection.

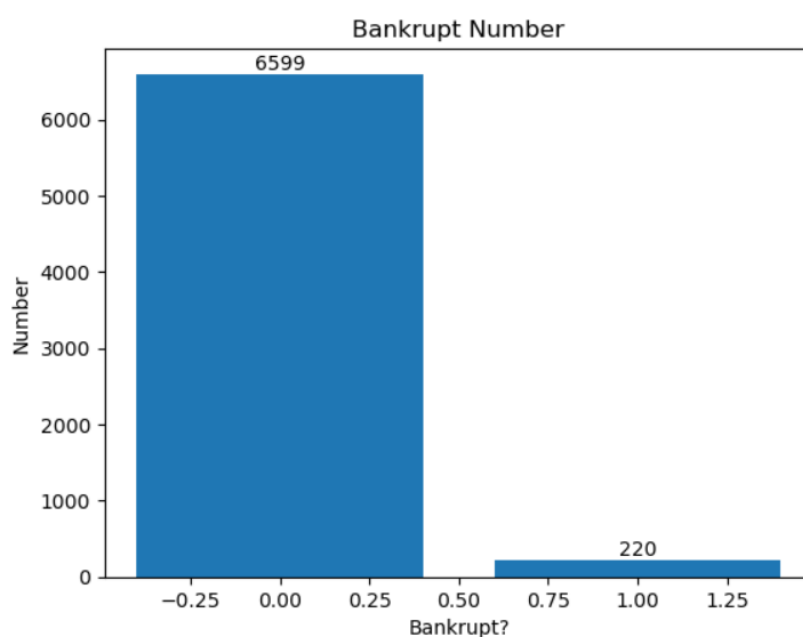


Figure 6: Target value's distribution (class imbalance)

A more dynamic approach to feature selection was also implemented using SHAP (Shapley Additive Explanations). Instead of relying solely on traditional statistical measures, SHAP values were leveraged to assess feature importance directly from the trained models. By analysing SHAP explanations, the most impactful financial indicators were identified. The effectiveness of SHAP-driven feature selection has been validated by **Chen et al. (2023)**, who

demonstrated that SHAP improves model transparency and prediction accuracy in high-dimensional financial datasets.

Class imbalance was another critical step in data processing, as the dataset exhibited in Figure 6. The original dataset contained 6,599 solvent firms and only 220 bankrupt firms, leading to a highly imbalanced distribution. Predictive models would be biased in favour of the majority class without intervention, making it difficult to detect businesses that are at risk of going bankrupt. The dataset was balanced by generating artificial samples of the minority class using the Synthetic Minority Over-Sampling Technique (SMOTE). The effectiveness of SMOTE in improving recall for financial distress prediction has been highlighted by **Smiti and Soui (2020)**, who demonstrated that oversampling techniques enhance the model's ability to detect bankrupt firms without compromising overall performance. After applying SMOTE, the training dataset achieved an equal distribution of bankrupt and non-bankrupt firms, allowing machine learning algorithms to learn from balanced data. The test dataset, however, retained its original class distribution to simulate real-world conditions where bankruptcies remain rare.

4.3 EXPLORATORY DATA ANALYSIS

An exploratory data analysis (EDA) is conducted in this part, considering two aspects: the dataset's 95 variables and a smaller collection of 7 features chosen based on the correlation matrix and ANOVA F-test findings. This analysis aims to identify patterns and relationships that could inform the development of our bankruptcy prediction model.



Figure 7: Distribution of all 95 variables of Bankruptcy dataset

Analysis of All Variables:

This approach involves visualizing the distribution of all 95 variables. The histograms (Figure 7) show a wide variation in the distributions of features, indicating that most variables are heavily skewed, and others follow a more normal distribution.

Analysis of Selected Features (Correlation Matrix and ANOVA F-test):

In this section, we explore seven selected features that were identified as significant based on the correlation matrix and ANOVA F-test. The analysis investigates the relationship between these features and bankruptcy status, as shown in the scatter plots.

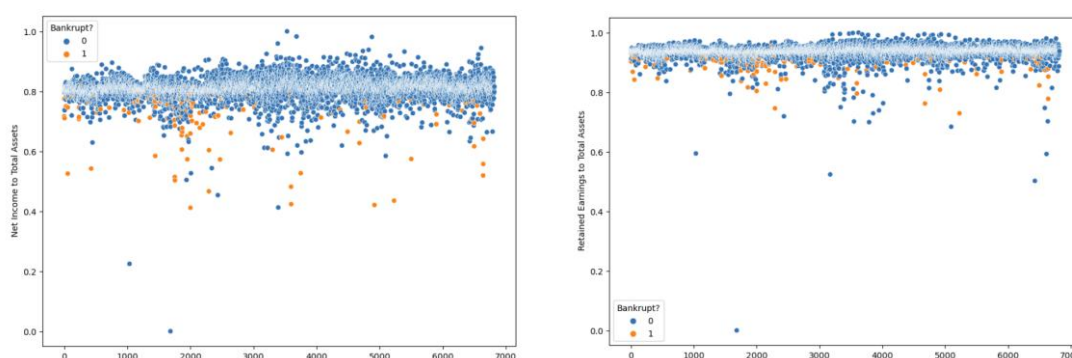


Figure 8: Distribution of Net Income & Retained Earnings

The first scatter plot (Figure 8) shows that most companies (represented by blue dots) have values clustered around 0.8, indicating that their net income to total assets are relatively high. In contrast, bankrupt companies (represented by orange dots) tend to have lower values for this ratio. This pattern aligns with earlier research by **Gaurav et al. (2024)**, which proposed that a lower net income as a percentage of total assets is a sign of financial difficulty.

Similar to the first feature (Figure 8), companies that have higher retained earnings (above 0.9) to total assets ratios generally exhibit financial stability (blue dots). However, bankrupt companies (orange dots) are concentrated in the lower range of this ratio.

Non-bankrupt businesses consistently have a net profit ratio above 0.2, according to the first scatter plot in Figure 9, but bankrupt businesses exhibit greater fluctuation and typically have lower values. This supports the findings of Arora & Saurabh (2022), who discovered that businesses are more likely to file for bankruptcy if their profit margin is low in comparison to their paid-in capital.

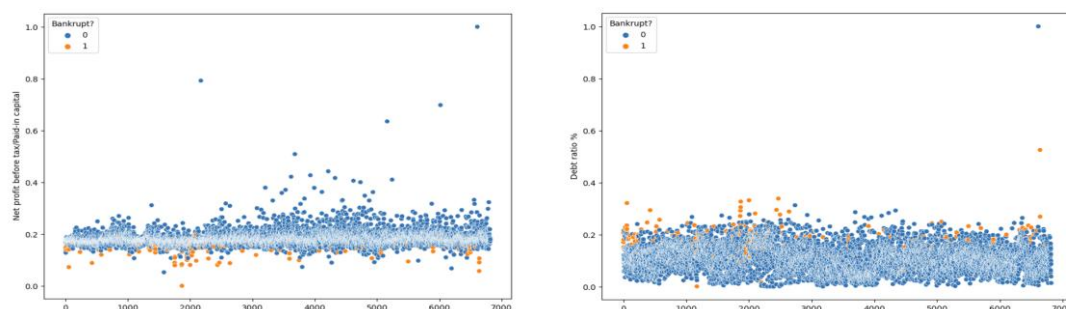


Figure 9: Distribution of Net Profit Before Tax to Paid-in Capital & Debt Ratio

In Figure 9, the debt ratio for bankrupt companies (orange dots) tends to be slightly higher (above 0.2) compared to non-bankrupt companies (blue dots). However, the difference is not as pronounced. This suggests that while debt might be a factor, it alone may not be as strong a predictor as the other financial ratios. **Liang et al. (2016)** also observed that debt ratios can influence bankruptcy predictions, but they are often not as decisive as other financial ratios.

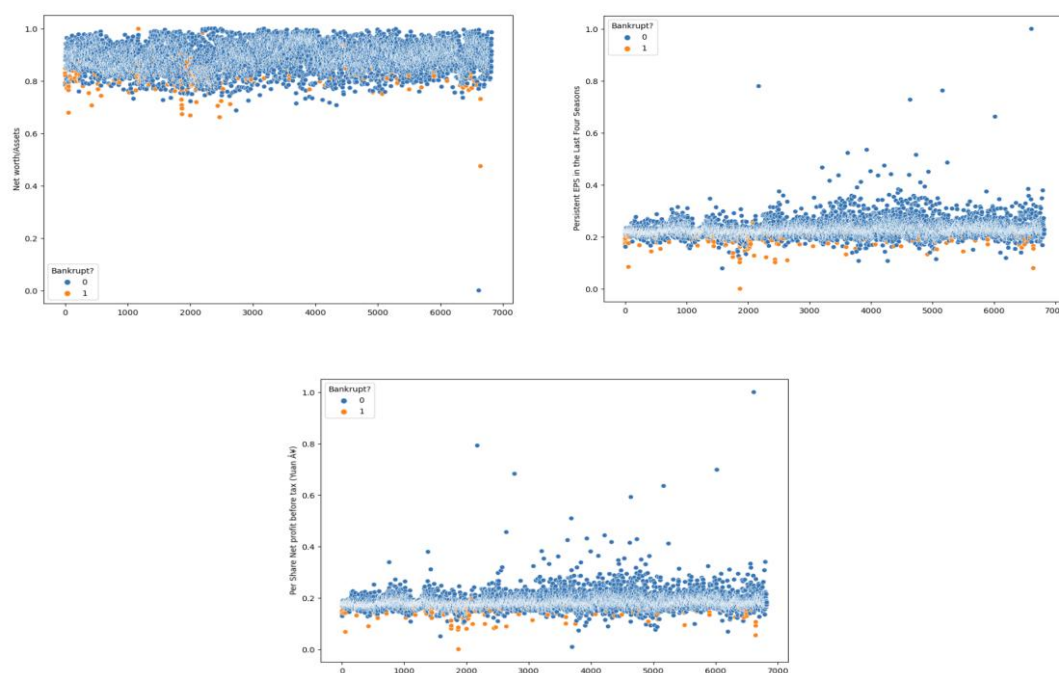


Figure 10: Distribution of Net Worth to Assets, Persistent EPS and
Per Share Net profit before Tax

While non-bankrupt enterprises are grouped around higher values (above 0.8), bankrupt companies (orange dots) display somewhat lower values for net worth to assets, as illustrated in Figure 10. Given that bad financial health is indicated by a lower net worth in relation to assets, this characteristic is closely associated with financial difficulty.

The second plot in Figure 10 shows that bankrupt companies often have lower (below 0.2) and more inconsistent values for earning per share, indicating that their EPS over the last four seasons have been unstable. **Lombardo et al. (2022)** emphasized that companies with persistent low or declining EPS over multiple periods are at a higher risk of bankruptcy.

The last scatter plot (Figure 10) shows that bankrupt companies generally have lower values (below 0.2) for net profit before tax per share. According to **Zhao et al. (2016)**, businesses with low profits per share are more likely to face financial difficulties since they have fewer resources available to pay their debts.

5 METHODOLOGY

In section 5, we use machine learning models to assess and contrast two different approaches for predicting bankruptcy. The first methodology involves feature selection, where 7 significant features are selected based on the correlation matrix and ANOVA F-test. The second methodology considers the full set of 95 features without any reduction. Three machine learning models - Random Forest, XGBoost, and LightGBM - are used to compare these two strategies.

5.1 RANDOM FOREST

5.1.1 HYPERPARAMETER TUNING

Hyperparameter tuning is the process of selecting the optimal set of model parameters to increase forecast accuracy and prevent overfitting. To do this, a predetermined set of hyperparameters is explored using Grid Search in order to determine which configuration produces the best model performance. After the tuning, the first methodology indicates the precision around 94.17%. For the second methodology, precision is improved to 97.39%, but the optimal hyperparameters remained similar, suggesting that the reduction in features slightly decreased model performance but potentially improved computational efficiency and interpretability. The research conducted by Barboza et al. (2017), which is in line with these findings, emphasized the significance of hyperparameter adjustment in enhancing the accuracy of bankruptcy prediction models.

5.1.2 EVALUATION MATRICES

The performance of two methodologies are evaluated based on Classification Report to provide insights into the trade-offs between using a reduced feature set and all available features. Other figures of evaluation matrices (confusion matrix, ROC AUC, PR AUC) are added in the Appendix.

	precision	recall	f1-score	support		precision	recall	f1-score	support
0	0.98	0.94	0.96	1980	0	0.98	0.97	0.98	1980
1	0.21	0.44	0.28	66	1	0.35	0.44	0.39	66
accuracy			0.93	2046	accuracy			0.96	2046
macro avg	0.59	0.69	0.62	2046	macro avg	0.67	0.71	0.68	2046
weighted avg	0.96	0.93	0.94	2046	weighted avg	0.96	0.96	0.96	2046

Figure 11: Random Forest Classification Report - selected features(left) & all features(right)

Adding more features improved the overall model accuracy from 93% to 96%, with precision for predicting bankruptcy cases increasing from 21% to 35% (Figure 11). However, recall for bankrupt firms remained unchanged at 0.44, indicating that many bankrupt cases were still missed. The F1-score for bankrupt firms increased from 0.28 to 0.39, suggesting that the full feature set provides a better balance between precision and recall. These results align with **Kim et al. (2021)** and **Dasilas & Rigani (2024)**, who highlight the intricacy of financial data and the difficulty of identifying bankruptcy as a result of class imbalance.

5.1.3 K-FOLD CROSS VALIDATION

Two methodologies for bankruptcy prediction are compared using K-fold Cross Validation. The comparison provides insights into how the number of features influences model stability, recall performance, and variability across different data splits. Other figures of K-fold Cross Validation are added in the Appendix.

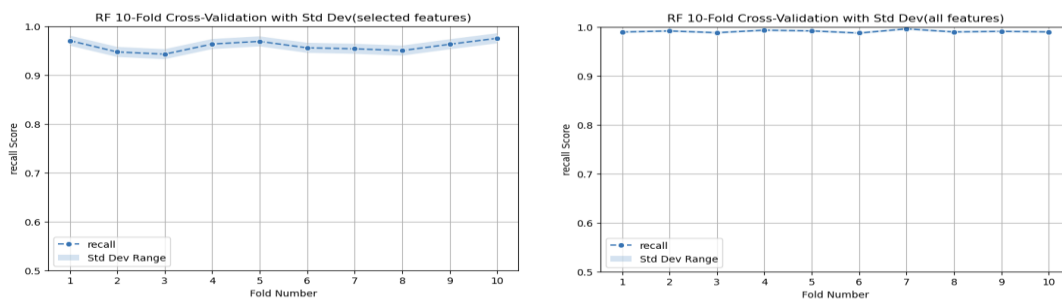


Figure 12: Random Forest K-fold CV - selected features(left) & all features(right)

As shown in Figure 12, the model using all features demonstrated significantly improved recall consistency, with scores approaching 1, indicating that the model's performance was more reliable for bankruptcy prediction. In contrast, the model with selected features exhibited more variation across folds, suggesting that predictions were less stable and more sensitive to the data splits.

5.1.4 MODEL EXPLAIN WITH SHAP

This section focuses on three key elements as we use SHAP to explain the Random Forest model: Instance-Level Explanation through the Force Plot, Global Feature Importance via the Summary Plot, and Feature Interactions through the SHAP Dependence Plot.

Instance-Level Explanation (Force Plot)

We examine the instance-level explanation for a specific company by comparing two methodologies. The model's rationale for its choice is made clear by the Force Plot, which shows how distinct characteristics affect the company's likelihood of going bankrupt in a particular case.

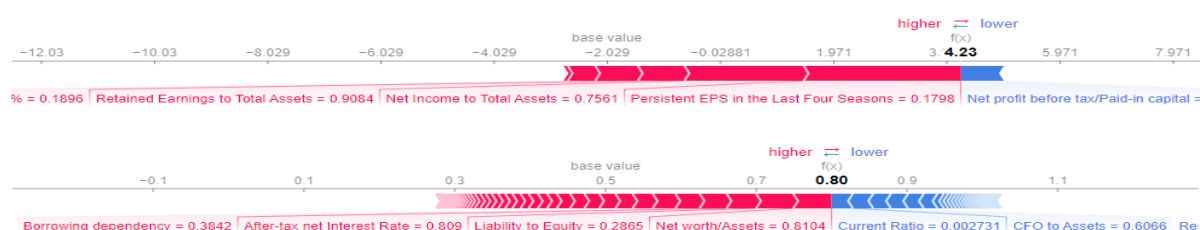


Figure 13: Random Forest Force Plot - selected features(first) & all features(second)

The red features dominate, leading to an increased bankruptcy risk prediction of 4.23, compared to the base value of -2.029 (Figure 13). Key risk indicators such as a high debt ratio, retained earnings, EPS, and net income all contribute to this higher risk, likely reflecting liquidity issues, reinvestment challenges, or financial sustainability concerns. However, the net profit before tax relative to paid-in capital acts as a mitigating factor, slightly reducing the bankruptcy risk, though not enough to outweigh the negative influences. In conclusion, the bankruptcy risk forecast is greatly increased by the dominating risk factors, such as excessive debt and poor profits, even though the moderating effect of net profit before taxes offers some relief (Kim et al., 2021).

The red features again dominate, pushing the bankruptcy risk prediction to 0.80, compared to the base value of 0.5 in Figure 13. Key risk factors such as high borrowing dependency, interest payments, and a high liability-to-equity ratio indicate excessive leverage, increasing financial instability. Although positive factors like the current ratio and CFO to assets ratio reduce the bankruptcy risk, their impact is insufficient to counteract the more significant negative factors. These results align with **Dasilas & Rigani (2024)**, who highlight the significant role of leverage and financial stability in bankruptcy prediction.

Global Feature Importance (Summary Plot)

This section provides an overview of each feature's contribution to the model's predictions. In both methodologies, five best features are selected, but the approach differs based on the number of features considered overall.

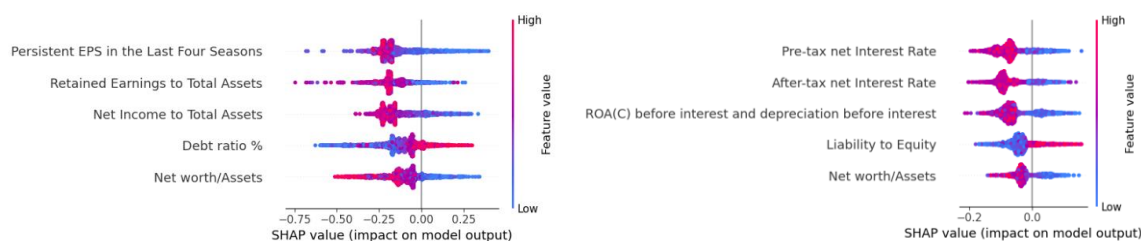


Figure 14: Random Forest Summary Plot - selected features(left) & all features(right)

High earnings stability, profitability, and retained earnings emerge as protective factors that reduce the probability of bankruptcy (Figure 14). High debt levels and weak net worth are identified as significant risk indicators contributing to financial distress. Notably, Persistent EPS and retained earnings are particularly influential, indicating that firms with strong profitability and consistent earnings tend to be more stable.

Most features exhibit a relatively small impact on bankruptcy risk, with SHAP values clustering around zero (Figure 14). However, pre-tax and after-tax net interest rates stand out as the most influential features affecting the model's predictions.

Feature Interactions (SHAP Dependence Plot)

The SHAP dependence plots illustrate how selected financial variables interact with other influential features, providing insights into their impact on bankruptcy prediction. The most crucial elements for dependence plots are thought to be the pre-tax net interest rate from the

second approach and the persistent EPS in the last four seasons from the first way. Further dependence plots for additional features are provided in the appendix.

It is evident in Figure 15 that when Persistent EPS is below 0.2, the bankruptcy risk increases, especially for firms with high debt ratios. As EPS exceeds 0.2, its impact on bankruptcy risk stabilizes, and firms with strong earnings persistence tend to experience reduced financial distress. Notably, firms with low debt ratios remain stable regardless of fluctuations in EPS, while those with high leverage are more vulnerable to financial instability. The analysis suggests that while strong earnings persistence helps mitigate bankruptcy risk (Kim et al. 2021), highly leveraged firms remain particularly susceptible to financial distress, emphasizing the critical role of debt management in financial stability.

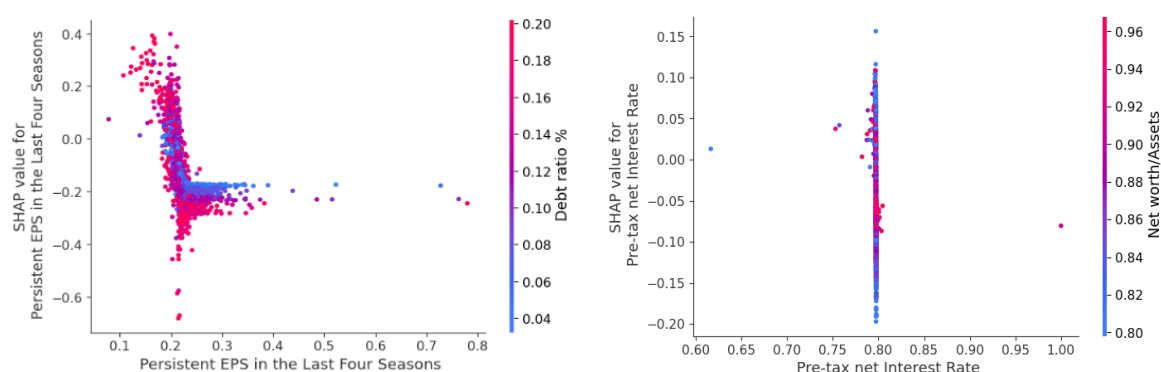


Figure 15: Random Forest Dependence Plot - selected features(left) & all features(right)

Pre-tax net interest rate has minimal impact on bankruptcy risk, as shown by the near-zero SHAP values (Figure 15). This is largely due to the fact that most firms in the dataset have similar pre-tax net interest rates (~ 0.8), which leads to limited variation in financial distress attributed to this feature. Additionally, net worth to assets does not significantly alter the effect of pre-tax net interest rates on bankruptcy risk, reinforcing the notion that leverage and interest rates play a limited role in this particular dataset.

5.2 XGBOOST

5.2.1 HYPERPARAMETER TUNING

After conducting hyperparameter tuning in XGBoost, the first methodology achieved a precision of approximately 92.50%, while the second methodology demonstrated an

improvement to 97.75%. The optimal hyperparameters remained largely consistent across both methodologies, with notable adjustments in max_depth, min_child_weight, and subsample.

5.2.2 EVALUATION MATRICES

The performance of two methodologies are evaluated in XGBoost based on Classification Report. Other figures of evaluation matrices are added in the Appendix.

	precision	recall	f1-score	support		precision	recall	f1-score	support
0	0.98	0.93	0.95	1980	0	0.98	0.97	0.98	1980
1	0.19	0.52	0.28	66	1	0.34	0.45	0.39	66
accuracy			0.91	2046	accuracy			0.95	2046
macro avg	0.59	0.72	0.61	2046	macro avg	0.66	0.71	0.68	2046
weighted avg	0.96	0.91	0.93	2046	weighted avg	0.96	0.95	0.96	2046

Figure 16: XGBoost Classification Report - selected features(left) & all features(right)

The model using all features in Figure 16 significantly improved precision from 0.19 to 0.34 and overall accuracy from 91% to 95%. This demonstrates that including a broader set of features enhances the model's capacity to accurately categorize non-bankrupt firms. However, while the selected features model achieved a higher recall (0.52) for bankrupt firms, it also led to more false positives.

5.2.3 K-FOLD CROSS VALIDATION

In this section, we compare two methodologies in XGBoost for bankruptcy prediction using K-fold Cross Validation. Other figures of validation are added in the Appendix.

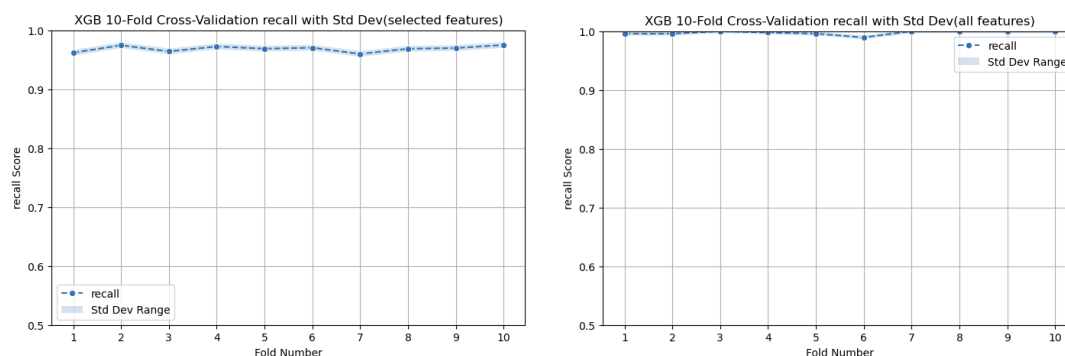


Figure 17: XGBoost K-fold CV - selected features(left) & all features(right)

The model using all features maintains nearly perfect recall (Figure 17), with values close to 1.0 and minimal variation. While this suggests strong model performance, it may indicate overfitting. In contrast, the model using selected features exhibits slight fluctuations in recall, suggesting better generalization and a more efficient model.

5.2.4 MODEL EXPLAIN WITH SHAP

Now, the XGBoost model is explained using SHAP focusing on three same key aspects: Instance-Level Explanation, Global Feature Importance, and Feature Interactions.

Instance-Level Explanation (Force Plot)

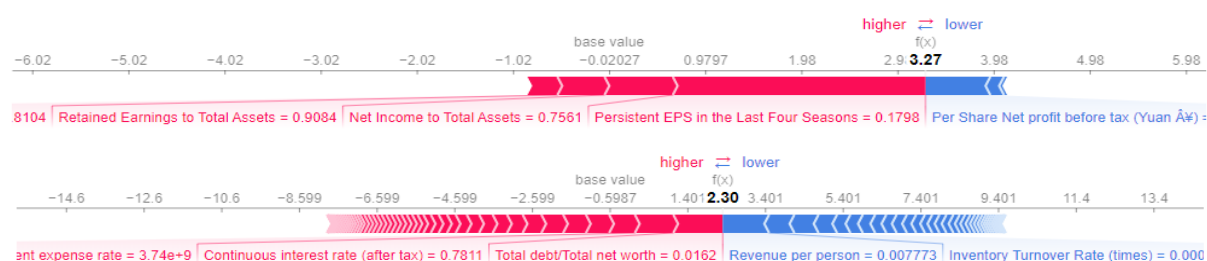


Figure 18: XGBoost Force Plot - selected features(first) & all features(second)

The red features dominate, leading to a higher bankruptcy risk prediction of 3.27, despite a near-zero base value of -0.02027 in Figure 18. Key risk indicators, such as high retained earnings and net income, increase the bankruptcy risk, possibly due to concerns about unsustainable earnings, poor reinvestment, or liquidity. The net profit before tax per share acts as a positive mitigating factor, slightly reducing the risk but not enough to offset the other negative influences.

The red features again dominate, pushing the bankruptcy risk prediction to 2.30, compared to the base value of -0.5987 (Figure 18). High payment expenses, interest rates, and debt levels significantly contribute to financial distress. However, positive mitigating factors like revenue per person and inventory turnover rate help reduce bankruptcy risk, but their impact is insufficient to counterbalance the negative factors.

Global Feature Importance (Summary Plot)

Key risk factors such as a high debt ratio and low net income dramatically raise the bankruptcy probability, as shown in Figure 19. Conversely, the likelihood of bankruptcy is decreased by

stability factors such as strong net income, net wealth, and retained earnings. As evidenced by its broad SHAP value distribution, Net Income to Total Assets is the most significant element, highlighting its significance in bankruptcy prediction.

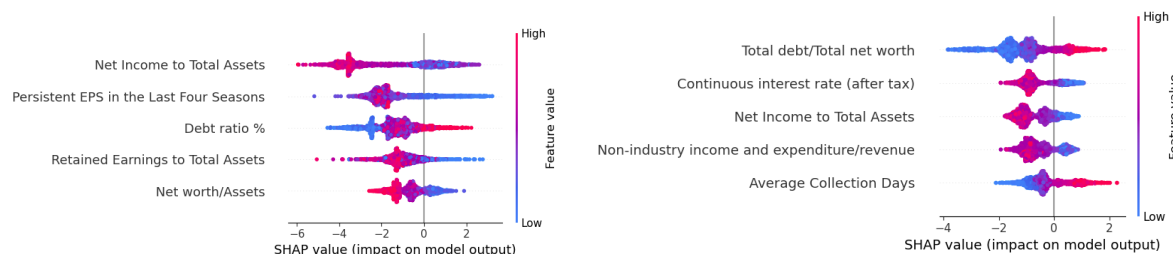


Figure 19: XGBoost Summary Plot - selected features(left) & all features(right)

Key risk indicators like high total debt to net worth and low net income to total assets strongly contribute to an increased bankruptcy risk (Figure 19). Conversely, higher net income relative to assets and higher continuous interest rates help mitigate bankruptcy risk. Debt-related factors, such as the total debt to net worth ratio, have the most significant impact, highlighting the importance of managing leverage effectively.

Feature Interactions (SHAP Dependence Plot)

The relationship between Net Income and Persistent EPS is shown in the first SHAP dependency plot in this section. Second plot illustrates the relationship between Total Debt and Net Income, showing how they influence bankruptcy risk.

When Net Income to Total Assets exceeds 0.8 in figure 20, bankruptcy risk is significantly reduced, with high Persistent EPS further reinforcing financial stability. A sharp transition in SHAP values around 0.8 indicates that firms below this threshold are more susceptible to financial distress. Firms with low Net Income to Total Assets and inconsistent EPS (represented by blue points) are at higher risk, reflecting unstable profitability.

Most firms with low Total Debt to Net Worth face lower bankruptcy risk (Figure 20), particularly when Net Income to Total Assets is high. However, a few highly leveraged firms exhibit varying bankruptcy risk depending on other factors. Firms with high Net Income to Total Assets are sometimes surprisingly vulnerable to financial distress because of their poor money management, as indicated by the wider variability in their SHAP values.

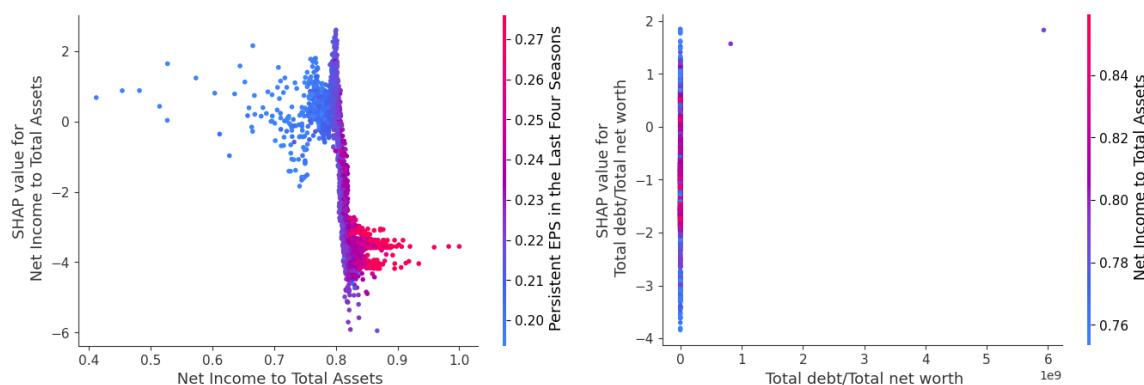


Figure 20: XGBoost Dependence Plot - selected features(left) & all features(right)

5.3 LIGHTGBM

5.3.1 HYPERPARAMETER TUNING

After tuning the hyperparameters in LightGBM, the first method, which used statistically selected features, achieved a precision of about 92.70%. In contrast, the second method, which included all features, showed an improvement, reaching 98.11% precision. Despite this increase in performance, the optimal hyperparameters remained mostly the same for both methods, with `colsample_bytree` and `num_leaves` being the main parameters that changed.

5.3.2 EVALUATION MATRICES

Based on the Classification Report, we compare the effectiveness of two approaches in LightGBM in this section. Other figures of evaluation matrices are added in the Appendix.

	precision	recall	f1-score	support		precision	recall	f1-score	support
0	0.98	0.93	0.96	1980	0	0.98	0.98	0.98	1980
1	0.21	0.55	0.30	66	1	0.43	0.45	0.44	66
accuracy			0.92	2046	accuracy			0.96	2046
macro avg	0.59	0.74	0.63	2046	macro avg	0.71	0.72	0.71	2046
weighted avg	0.96	0.92	0.93	2046	weighted avg	0.96	0.96	0.96	2046

Figure 21: Classification Report - selected features(left) & all features(right)

The model using all features significantly improved precision (Figure 21), rising from 0.21 to 0.43, and overall accuracy increased from 92% to 96%. This improvement suggests that the model with more features is more effective for real-world applications. However, while the selected features model achieved a higher recall for bankrupt firms (0.55), it also resulted in more false positives, which may not be ideal for financial risk assessment. These findings align

with **Mai et al. (2018)**, **Chen et al. (2020)**, and **Hosaka (2019)**, who note that feature selection impacts recall, but hybrid models combining machine learning and deep learning techniques can further enhance bankruptcy prediction accuracy.

5.3.3 K-FOLD CROSS VALIDATION

Two methodologies in XGBoost for bankruptcy prediction are evaluated in this section using K-fold Cross Validation. Other figures of validation are added in the Appendix.

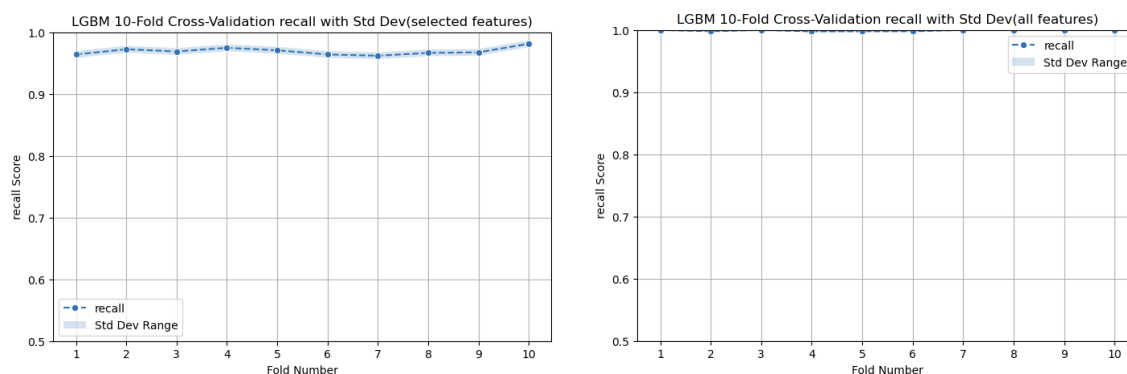


Figure 22: LightGBM K-fold CV - selected features(left) & all features(right)

The model using all features achieves perfect recall (1.0) across all folds in Figure 22. While this indicates strong performance, it may also suggest overfitting, as the model relies heavily on specific features. In contrast, the model with selected features maintains high recall with minor fluctuations, indicating better generalization while keeping the model efficient.

5.3.4 MODEL EXPLAIN WITH SHAP

The LightGBM model is explained in this part using SHAP, with a focus on three key areas: feature interactions, global feature importance, and instance-level explanation.

Instance-Level Explanation (Force Plot)

Positive SHAP contributions (in red) dominate, leading to a final prediction of 4.23, indicating a higher probability of bankruptcy in Figure 23. Key contributors to financial distress in this case are high retained earnings, net income, and earnings per share. While net profit slightly mitigates risk, its effect is not strong enough to outweigh the negative influences.

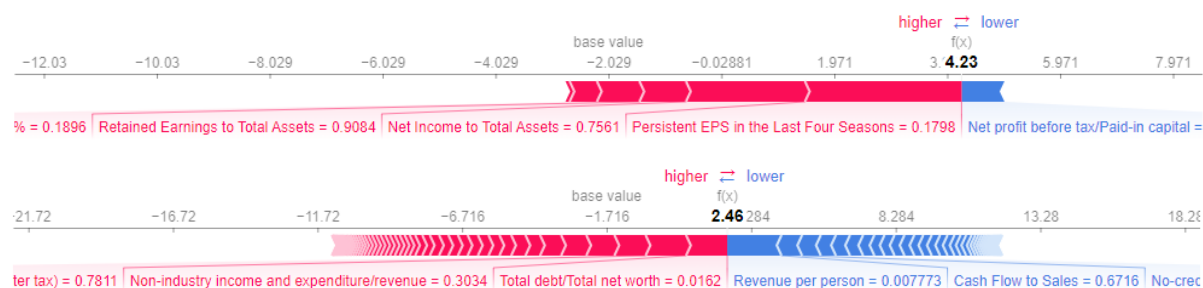


Figure 23: LightGBM Force Plot - selected features(first) & all features(second)

The red factors again dominate, pushing the bankruptcy risk higher to 2.46, compared to the base value of -1.716 (Figure 23). The company's reliance on non-core income and high debt levels significantly increases its financial risk. However, positive factors like cash flow and revenue efficiency help mitigate the bankruptcy risk, though their impact is not sufficient to counterbalance the negative influences.

Global Feature Importance (Summary Plot)

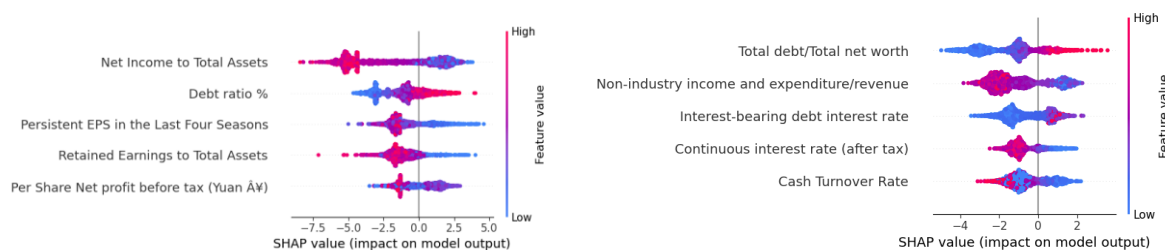


Figure 24: LightGBM Summary Plot - selected features(left) & all features(right)

High debt ratio, low net income, and low retained earnings emerge as significant risk indicators in Figure 24, all contributing to an increased bankruptcy risk. Conversely, higher net income, retained earnings, and per-share net profit act as stability indicators, reducing the likelihood of bankruptcy. Debt ratio and net income are identified as the most influential factors, highlighting the importance of effective debt management and sustained income for financial health.

High total debt to net worth ratio and low continuous interest rates are key risk factors that increase bankruptcy risk (Figure 24). Stability indicators such as higher non-industry income and cash turnover rate help mitigate this risk, underscoring the importance of managing debt levels and maintaining a strong cash flow.

Feature Interactions (SHAP Dependence Plot)

Net Income to Total Assets above 0.8 significantly reduces bankruptcy risk in Figure 25, especially when combined with high Net Profit Before Tax/Paid-in Capital. A sharp transition around 0.8 suggests that firms with lower net income need additional financial strengths to mitigate bankruptcy risk. It is often correlating with lower Net Profit Before Tax/Paid-in Capital, making them more vulnerable to financial distress.

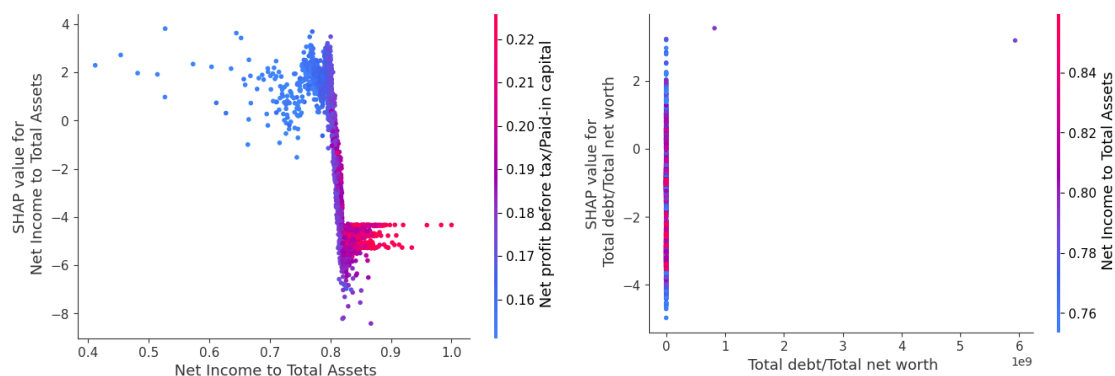


Figure 25: LightGBM Dependence Plot - selected features(left) & all features(right)

Most firms with low total debt face reduced bankruptcy risk, particularly when Net Income to Total Assets is high (Figure 25). However, highly leveraged firms show unclear impacts, likely due to sparse data and mixed SHAP values. Firms with low Net Income to Total Assets are more vulnerable to debt-related financial distress, as indicated by the increased variability in their SHAP values (Chen et al., 2020).

6 THESIS DISCUSSION

This chapter delves into the results obtained from the machine learning models (Random Forest, XGBoost, and LightGBM) and their comparative performance using two feature selection strategies: traditional methods (ANOVA F-tests & correlation matrices) and a SHAP-based dynamic feature importance approach. By linking the findings to existing literature and classical bankruptcy prediction methods, this discussion highlights the key insights, model limitations, and contributions of this research.

6.1 FEATURE SELECTION

Feature selection plays a crucial role in improving model presentation by classifying the most relevant financial indicators. Traditional methods of feature selection, such as ANOVA F-tests and correlation matrices, have long been employed in bankruptcy prediction to reduce dimensionality and eliminate redundant variables. These methods are grounded in statistical techniques and emphasize the importance of selecting features based on their statistical significance and correlation with the target variable.

Previous studies, including **Barboza et al. (2017)** and **Shetty et al. (2022)**, have highlighted the effectiveness of such traditional feature selection techniques in bankruptcy prediction. For example, **Barboza et al. (2017)** showed that using well-chosen financial ratios could significantly improve model accuracy. In this study, applying ANOVA F-tests and correlation matrices helped to identify key financial ratios that most effectively predict bankruptcy. Key features such as Net Income, Debt Ratio, and Retained Earnings were found to be most significant in the traditional feature selection process, aligning with previous research.

However, while these traditional techniques are effective, they do not capture complex, non-linear relationships between features. This limitation is addressed by the second feature selection approach, which utilizes SHAP to determine feature importance during model training. SHAP offers a more dynamic and model-driven feature selection method, which has become increasingly popular due to its ability to explain the impact of each attribute to model predictions. In contrast to static statistical approaches, SHAP allows for the exploration of feature interactions and the identification of non-linear relationships. This research found that SHAP provided deeper insights into the model's reasoning, particularly regarding complex relationships among financial indicators, thus overcoming the limitations of traditional feature selection.

The trade-offs between predictive performance and computational efficiency are highlighted by the contrast between SHAP and conventional feature selection. While traditional methods are more computationally efficient and easier to implement, SHAP-driven selection significantly improves model transparency and captures more nuanced feature interactions. This dynamic approach proves particularly useful in bankruptcy prediction, where non-linear dependencies between financial ratios are common.

6.2 BANKRUPTCY PREDICTION MODEL

The evaluation of the models under both methodologies offers a comprehensive understanding of how feature selection affects predictive performance and highlights the strengths and weaknesses of each model.

Model	Precision (First Methodology)	Precision (All Features)	Performance Change
Random Forest	94.17%	97.39%	+3.22%
XGBoost	92.50%	97.75%	+5.25%
LightGBM	92.70%	98.11%	+5.41%

Table 01: Hyperparameter Tuning Comparison

When using every feature instead of just the chosen subset, all three models perform better (Table 01). The results of earlier studies (**Barboza et al., 2017**) indicate that the model's capacity to identify intricate patterns in the financial data is generally improved by the inclusion of additional attributes. Among the models, LightGBM shows the most significant precision improvement (+5.41%) when all features are used, implying that LightGBM is more sensitive to the addition of features. XGBoost follows with a 5.25% improvement, while Random Forest shows a relatively modest improvement of 3.22%. This suggests that Random Forest may not be as reliant on a larger set of features as XGBoost and LightGBM, which could imply that the Random Forest is robust in handling a smaller, well-curated set of features. The findings are consistent with the research of **Chen et al. (2020)** and **Liang et al. (2016)**, who have demonstrated that gradient-boosting models like XGBoost and LightGBM are better at identifying intricate, non-linear relations in data, thus benefiting from a larger feature set.

Model	Precision (Selected Features)	Precision (All Features)	Recall (Selected Features)	Recall (All Features)
Random Forest	0.98	0.98	0.94	0.97
XGBoost	0.98	0.98	0.93	0.97
LightGBM	0.98	0.98	0.93	0.98

Table 02: Evaluation Matrices Comparison (Non-Bankrupt Companies)

The precision remains unchanged across all models when distinguishing non-bankrupt companies in Table 02, indicating that the models can confidently classify non-bankrupt firms

with minimal false positives. This finding supports the notion that machine learning models are effective in predicting solvent companies, as bankruptcy is relatively rare in the dataset, mirroring real-world conditions where most companies remain solvent.

However, Recall improves when all features are used. LightGBM demonstrates the highest recall of 0.98 with all features, highlighting its superior sensitivity in detecting non-bankrupt firms. This also suggests that LightGBM is particularly adept at identifying the majority class, a crucial feature in imbalanced datasets. Similar results were obtained by **Radwan et al. (2022)**, who also discovered that LightGBM is quite effective at classifying the majority class.

Model	Precision (Selected Features)	Precision (All Features)	Recall (Selected Features)	Recall (All Features)
Random Forest	0.21	0.35	0.44	0.44
XGBoost	0.19	0.34	0.52	0.45
LightGBM	0.21	0.43	0.55	0.45

Table 03: Evaluation Matrices Comparison (Bankrupt Companies)

For bankrupt companies, as shown in Table 03, the performance metrics suggest that precision improves across all models when all features are included. This suggests that the models are more confident in their bankruptcy predictions when given access to more information. LightGBM outperforms the other models, with the highest precision of 0.43 and the highest recall (0.55) when using all features. However, it is worth noting that XGBoost and LightGBM experience a slight decrease in recall with all features, suggesting that while adding features improves precision, it may also lead to a slight reduction in sensitivity to detecting bankrupt firms. This trade-off between precision and recall is an important aspect to consider in bankruptcy prediction models. In contrast, Random Forest exhibits a stable recall but improves its precision, making it more reliable in balancing the trade-off between false positives and false negatives, as it maintains strong performance in both detecting bankrupt and non-bankrupt firms.

Further insights into model stability are provided in Table 04. Both XGBoost and LightGBM perform exceptionally well with all features, achieving perfect recall. This illustrates how adding more financial variables greatly increases the sensitivity of the model, making it better able to detect both bankrupt and non-bankrupt businesses. LightGBM stands out for its minimal standard deviation, indicating that it is the most stable model, irrespective of the feature set

used. This aligns with previous studies, such as **Ke et al. (2017)**, which emphasize the robustness of LightGBM in various financial prediction tasks, especially in datasets with imbalanced classes.

Model	Recall (Selected Features)	Recall (All Features)	Std Dev (Selected Features)	Std Dev (All Features)
Random Forest	0.94	1.0	0.02	0.0001
XGBoost	0.96	1.0	0.015	0.0001
LightGBM	0.97	1.0	0.01	0.0001

Table 04: K-fold Cross Validation Comparison

6.3 EXPLAINABLE AI

The integration of Explainable AI (XAI), particularly through the use of SHAP (Shapley Additive Explanations), provides valuable insights into the underlying factors that drive the bankruptcy predictions in this study. We can better understand how each financial feature influences the model's decision-making process by using SHAP, which increases transparency and fosters confidence in the machine learning forecasts. This section discusses the findings related to feature importance and the shift in focus from profitability to debt ratios across the models, highlighting the model interpretability in bankruptcy prediction.

Model	Selected Features - Best Feature	All Features - Best Feature
Random Forest	Persistent EPS in the Last Four Seasons	Pre-tax Net Interest Rate
XGBoost	Net Income to Total Assets	Total Debt/Total Net Worth
LightGBM	Net Income to Total Assets	Total Debt/Total Net Worth

Table 05: SHAP Global Feature Importance Comparison

Global Feature Importance derived from SHAP values reveals crucial insights into the financial indicators that drive bankruptcy predictions (Table 05). The analysis demonstrates that profitability features, such as Net Income is the most important in XGBoost and LightGBM when using the selected features. These results are consistent with findings in the literature, where profitability ratios have been frequently identified as strong predictors of bankruptcy (**Barboza et al., 2017**). Net income's dominance over total assets in both XGBoost and

LightGBM supports the idea that less profitable businesses are more prone to face financial difficulties, a key factor in bankruptcy predictions (**Zhao et al., 2016**).

Conversely, when all features are considered, the importance of debt ratios rises significantly in both XGBoost and LightGBM, with Total Debt/Total Net Worth ranking as the highest contributing factor. This shift towards debt ratios underscores the importance of financial leverage in bankruptcy prediction. The higher the debt levels in relation to a company's net worth, the more likely it is to face insolvency, especially when cash flows are insufficient to meet debt obligations. This is consistent with traditional models, which often place significant weight on leverage measures such as debt-to-equity and debt-to-assets ratios (**Altman, 1968**).

The dominance of debt ratios when all features are used suggests that XGBoost and LightGBM are effectively capturing the complex relationships between financial leverage and bankruptcy risk, which are often non-linear in nature. This finding supports previous research that highlights the superior ability of machine learning models, particularly gradient boosting algorithms, to capture these non-linear relationships (**Chen et al., 2020; Radwan et al., 2022**).

Random Forest and Shifting Feature Importance

For Random Forest, the most important feature shifts from earnings stability (measured by Persistent EPS in the Last Four Seasons) to interest expenses when all features are used. This shift suggests that while earnings stability remains a crucial indicator of financial health, interest expenses become more important when additional features are incorporated into the model. The emphasis on interest expenses highlights the model's sensitivity to financial costs and their impact on bankruptcy risk. This shift is consistent with the literature that emphasizes the critical role of managing financial costs in maintaining solvency (**Liang et al., 2016**). The sensitivity to interest expenses further supports the notion that effective financial management, especially with regard to debt servicing, is critical for a firm's survival.

In conclusion, this research not only reinforces the relevance of classical bankruptcy predictors but also introduces innovative approaches to feature selection and model interpretation. By combining traditional methods with SHAP-driven dynamic feature selection and evaluating machine learning models in depth, this study advances the field of bankruptcy prediction. It advances a more sophisticated comprehension of the connections between bankruptcy risk and

financial characteristics, laying the groundwork for further studies that combine machine learning and traditional financial theory.

7 PRACTICAL IMPLICATION

The practise of machine learning models, particularly Random Forest, XGBoost, and LightGBM, in bankruptcy prediction has shown significant promise but also comes with certain restrictions that essential to be addressed for practical applications.

One of the prime challenges with these models is their potential to overfit, especially when all features are used. As shown in the evaluation matrices and K-fold cross validation, while adding more features can improve recall and precision, it may also lead to models that rely too heavily on specific features, thus reducing their generalizability. For example, in the case of LightGBM, the model showed near-perfect recall when using all features, which might indicate overfitting, limiting its capability to simplify to novel, unseen data. Additionally, while these models perform better than traditional bankruptcy prediction models (such as Z-scores or financial ratio analysis), they can be computationally expensive, particularly for smaller businesses with less data, and they need a lot of data for training.

Conventional techniques for predicting bankruptcy, like Ohlson's O-score or Altman's Z-score, mostly depend on financial ratios and assume that factors have linear correlations with one another. These classical models, while useful, often miss out on the complex, non-linear interactions that machine learning models can capture. A particular advantage of the machine learning algorithms analysed in this study, such as XGBoost and LightGBM, is their capacity to recognize these complex connections between financial metrics, including debt ratios, liquidity, and profitability. However, traditional methods provide transparency and ease of understanding, which can be beneficial for regulatory compliance and decision-making in small businesses or non-technical settings. Therefore, a hybrid approach, combining both machine learning models and Explainable AI, could offer more comprehensive insights, improving the balance between predictive power and interpretability.

These machine learning algorithms help improve financial health monitoring and credit risk assessment in corporate analytics. They are particularly valuable in predicting bankruptcy risk, especially for firms with complex financial profiles, where traditional methods may fall short.

Models like XGBoost and LightGBM can offer deeper insights into financial hardship by capturing non-linear correlations and interactions between numerous variables, assisting firms in proactively addressing possible risks. For example, these models can be used by banks, investment firms, and credit rating agencies to assess the risk of lending or investment decisions.

From an economic perspective, this model can assist policymakers and regulators in identifying systemic risks within industries or economies. For instance, by predicting bankruptcy risk across sectors, they can highlight areas where economic intervention may be necessary to stabilize the market or prevent a financial crisis. The model can also be used for stress testing in economic research, where they simulate how various financial indicators, such as debt levels and profitability, impact the likelihood of firm failure during economic downturns or market shocks.

8 CONCLUSION

In order to anticipate financial insolvency, this thesis examined the usage of machine learning models, particularly Random Forest, XGBoost, and LightGBM. By integrating feature selection, model training, and Explainable AI (XAI) techniques, the study aimed to enhance bankruptcy forecasting, providing both predictive accuracy and model transparency. The findings demonstrate that machine learning models, especially when combined with SHAP-driven feature importance, can significantly improve bankruptcy risk assessment by uncovering complex, non-linear relationships between financial variables.

The research addressed key challenges faced in bankruptcy prediction, including class imbalance, feature selection, and model interpretability. Through traditional feature selection methods such as ANOVA F-tests and correlation matrices, the study identified important financial indicators for bankruptcy prediction. However, the integration of SHAP allowed for a more dynamic and transparent approach to feature importance, enabling the models to capture deeper insights into financial data interactions and improve interpretability.

In terms of model performance, the study found that while all three machine learning models showed improved predictive accuracy compared to traditional bankruptcy models like Altman's Z-score, they also posed challenges related to overfitting, particularly when using all

features. LightGBM, for instance, achieved near-perfect recall with all features, but this came at the cost of potential overfitting. The selected-feature models showed more stable recall scores, suggesting better generalization and efficiency.

In a nutshell this thesis uses explainable AI and machine learning to develop the expanding subject of financial risk calculation. The integration of feature selection, model training, and SHAP - driven explanations offers a scalable and interpretable approach to bankruptcy prediction. The outcomes of this study not only enhance the evaluation of bankruptcy risk but also offer a strong foundation for future financial decision-making, the creation of policies, and additional study in the ground of financial analytics and machine learning applications.

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APPENDIX

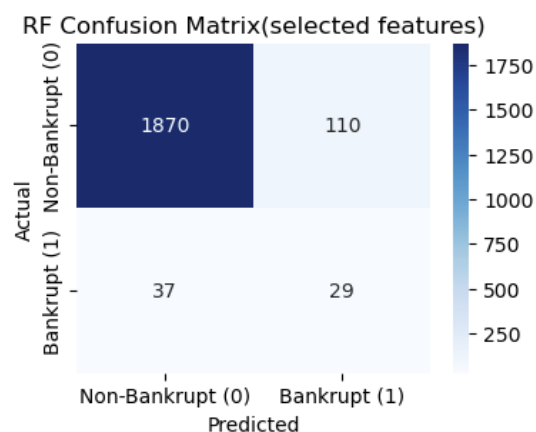


Figure 01: Random Forest Confusion Matrix (selected features)

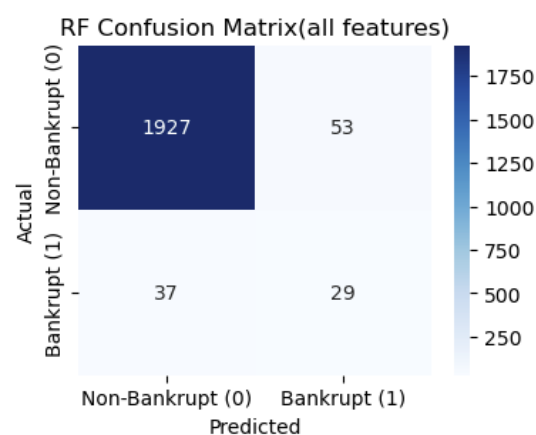


Figure 02: Random Forest Confusion Matrix (all features)

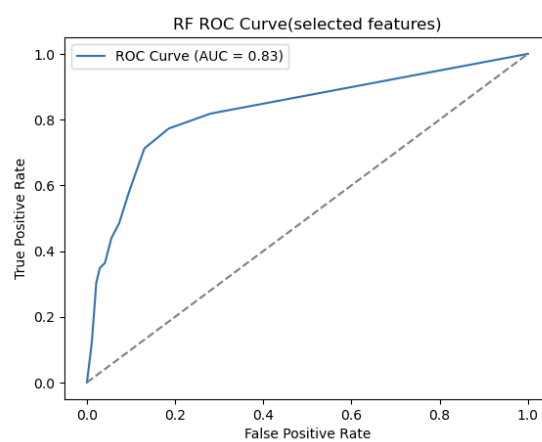


Figure 03: Random Forest ROC-AUC curve (selected features)

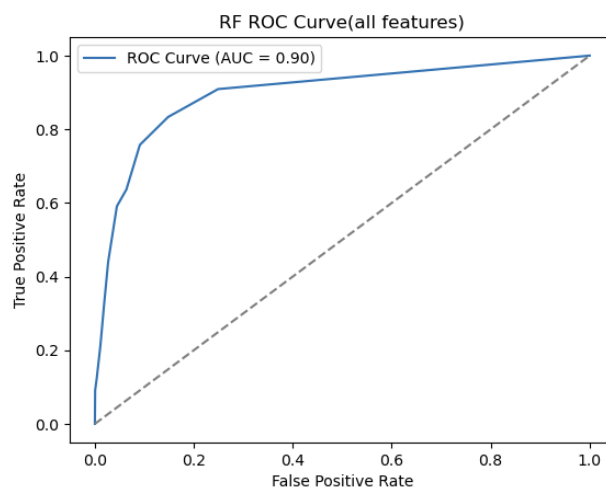


Figure 04: Random Forest ROC-AUC curve (all features)

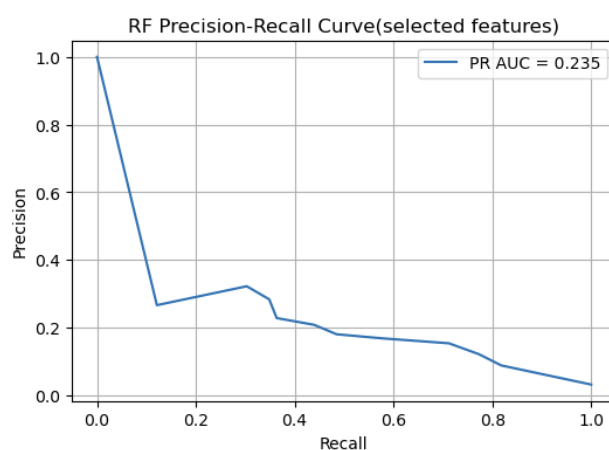


Figure 05: Random Forest PR-AUC curve (selected features)

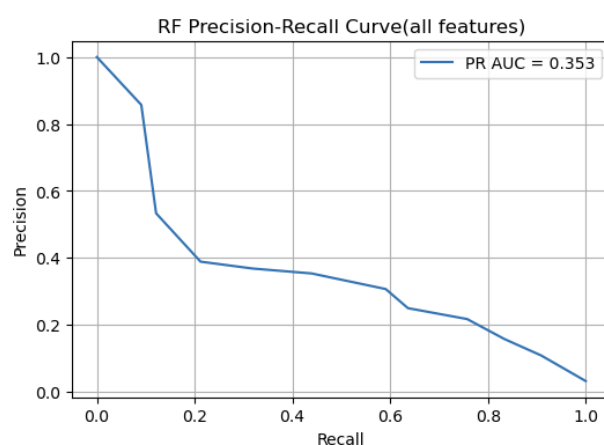


Figure 06: Random Forest PR-AUC curve (all features)

RF Distribution of Cross-Validation recall Scores(selected features)

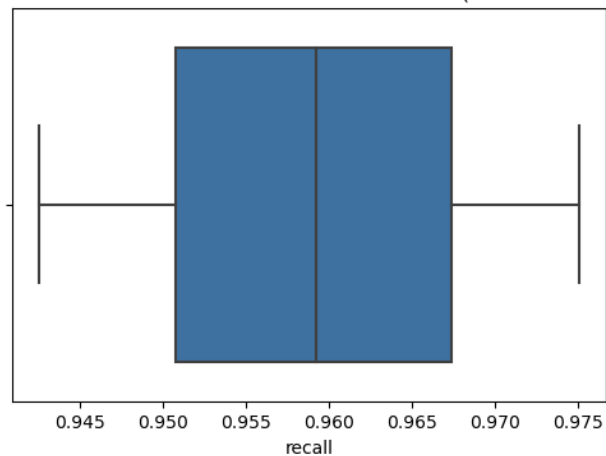


Figure 07: Random Forest Distribution of Cross-Validation recall scores (selected features)

RF Distribution of Cross-Validation recall Scores(all features)

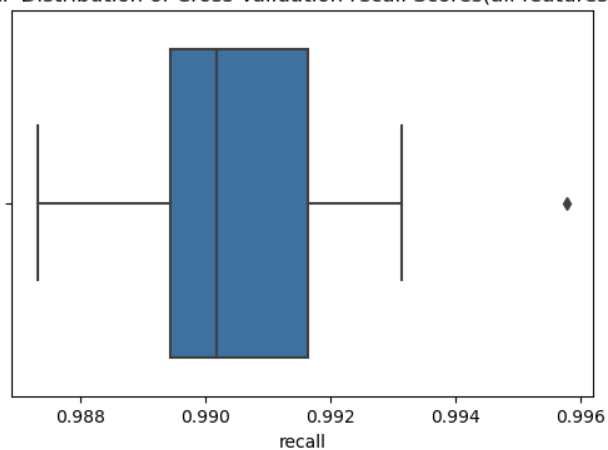


Figure 08: Random Forest Distribution of Cross-Validation recall scores (all features)

RF Histogram of k-Fold recall Scores(selected features)

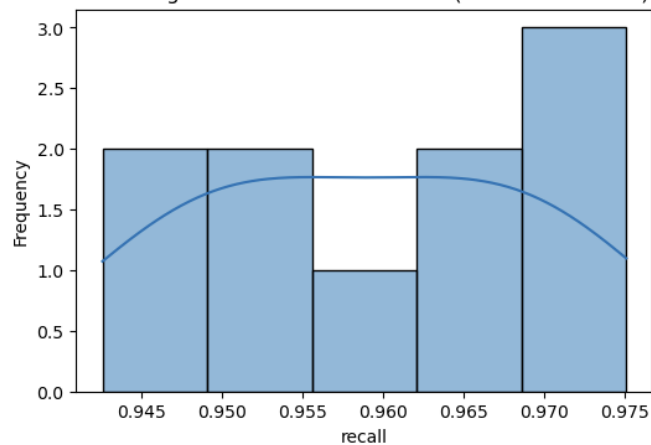


Figure 09: Random Forest Histogram of Cross-Validation recall scores (selected features)

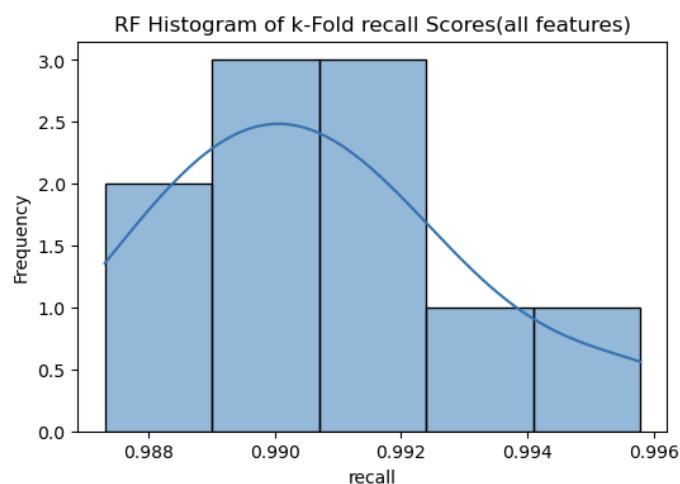


Figure 10: Random Forest Histogram of Cross-Validation recall scores (all features)

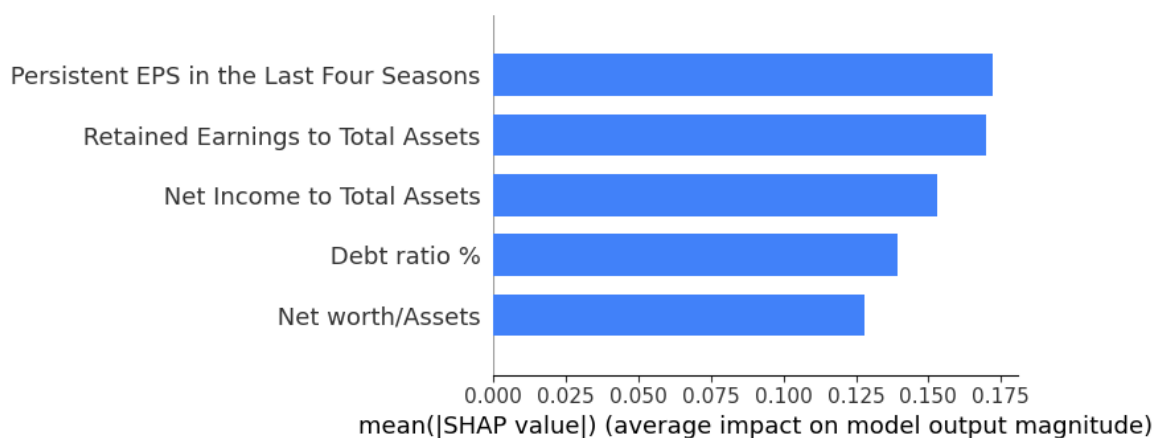


Figure 11: Random Forest SHAP Summary Plot (selected features)

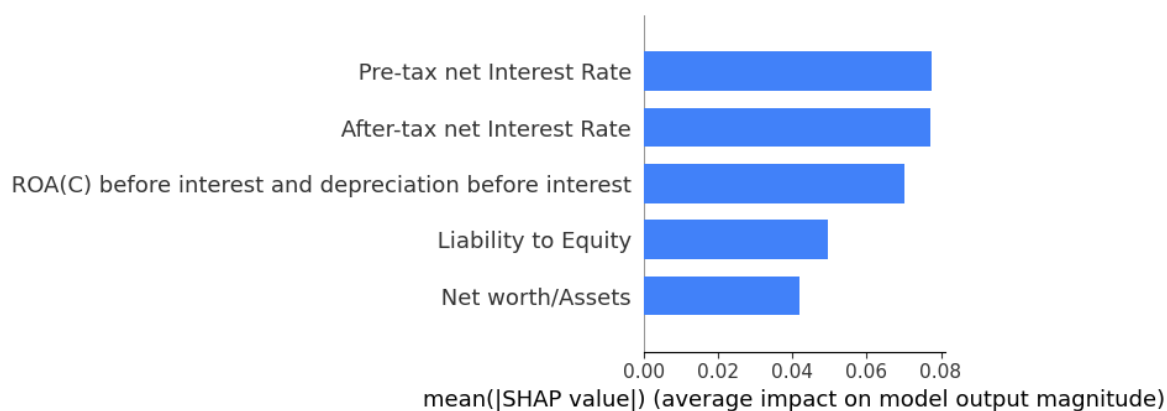


Figure 12: Random Forest SHAP Summary Plot (all features)

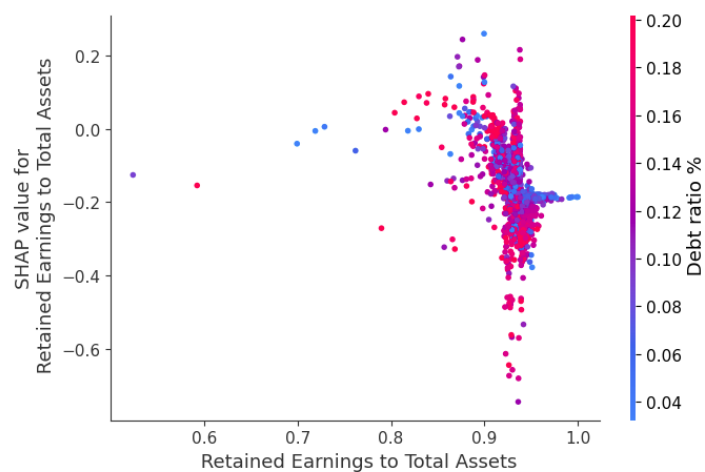


Figure 13: SHAP Dependence Plot (selected features)

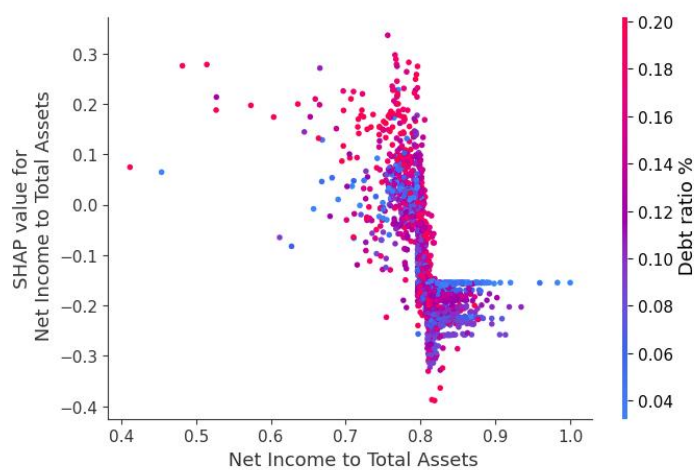


Figure 14: SHAP Dependence Plot (selected features)

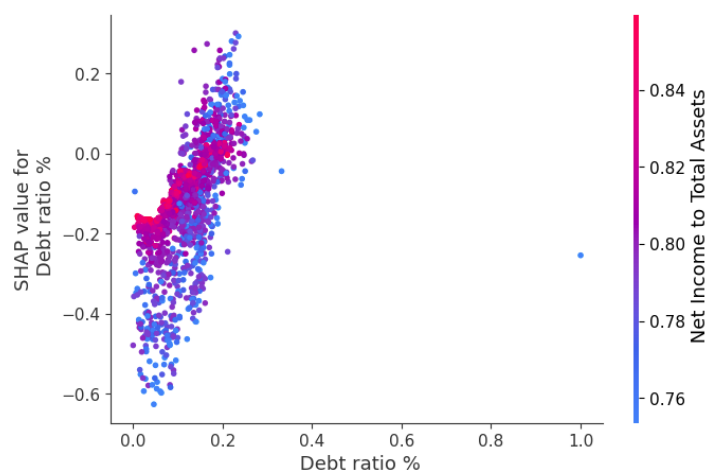


Figure 15: SHAP Dependence Plot (selected features)

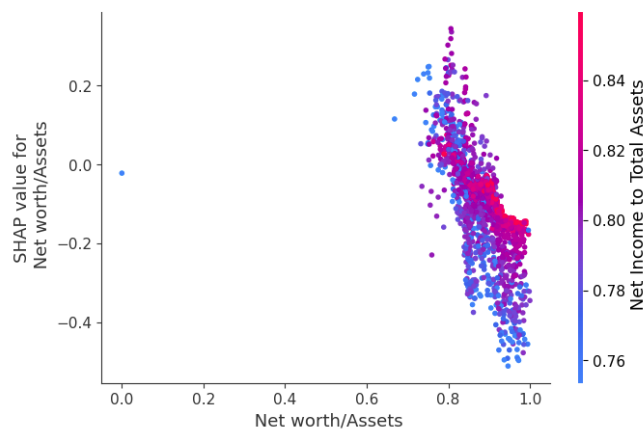


Figure 16: SHAP Dependence Plot (selected features)

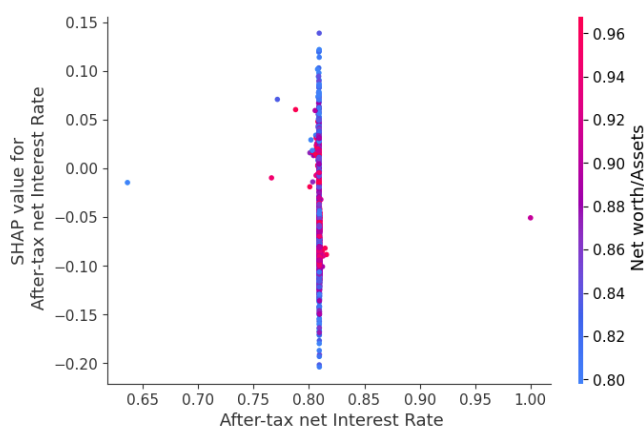


Figure 17: SHAP Dependence Plot (all features)

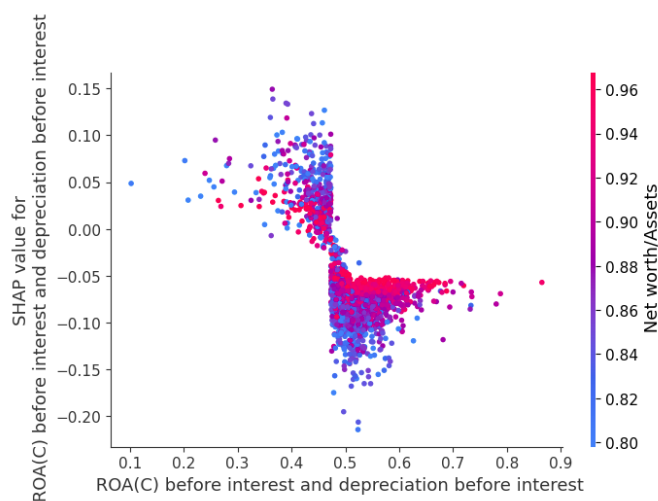


Figure 18: SHAP Dependence Plot (all features)

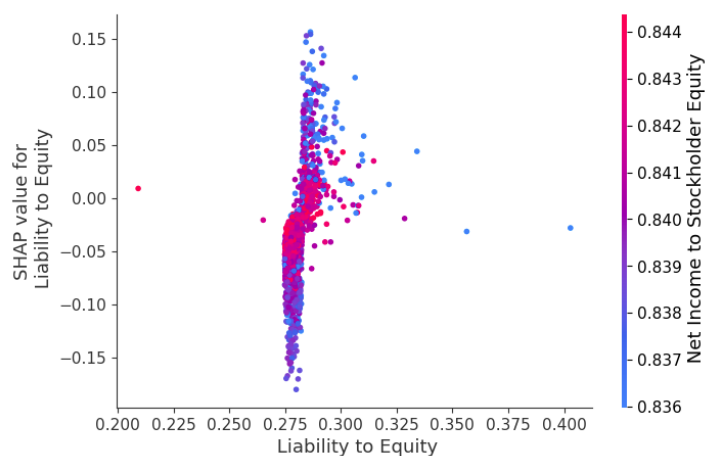


Figure 19: SHAP Dependence Plot (all features)

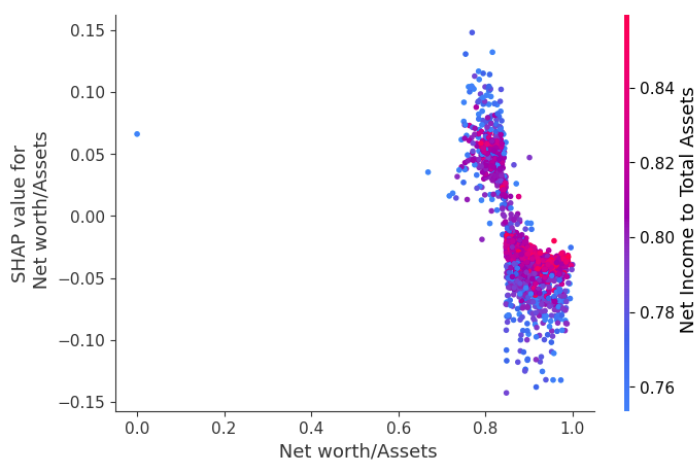


Figure 20: SHAP Dependence Plot (all features)

XGBoost Confusion Matrix(selected features)

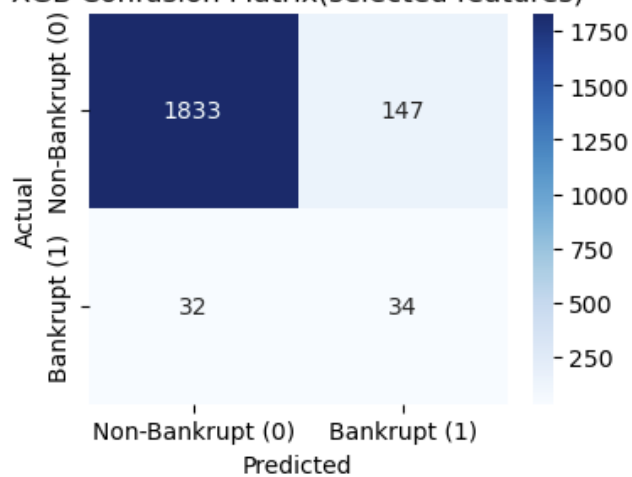


Figure 21: XGBoost Confusion Matrix (selected features)

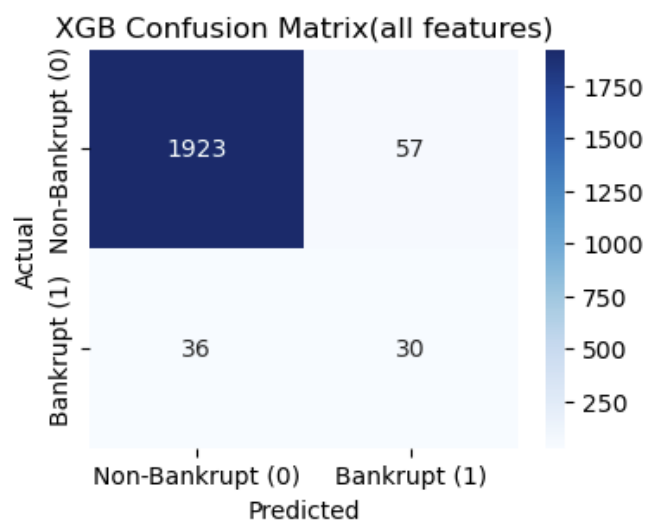


Figure 22: XGBoost Confusion Matrix (all features)

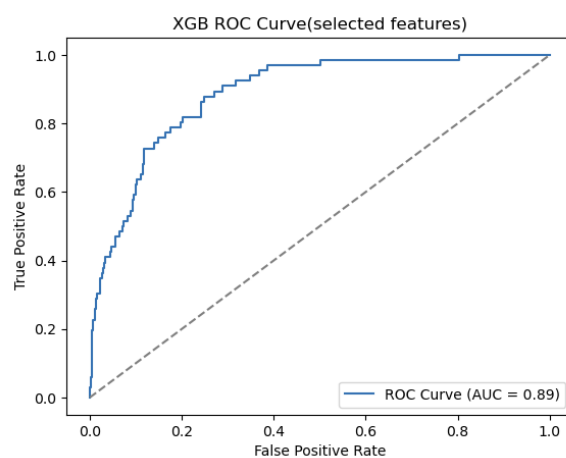


Figure 23: XGBoost ROC-AUC Curve (selected features)

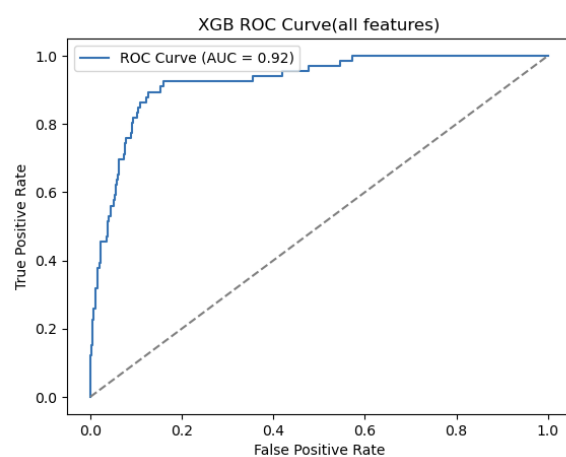


Figure 24: XGBoost ROC-AUC Curve (all features)

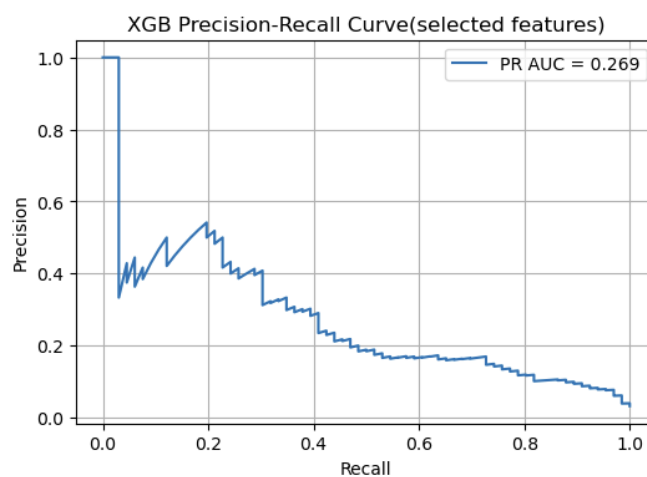


Figure 25: XGBoost PR-AUC Curve (selected features)

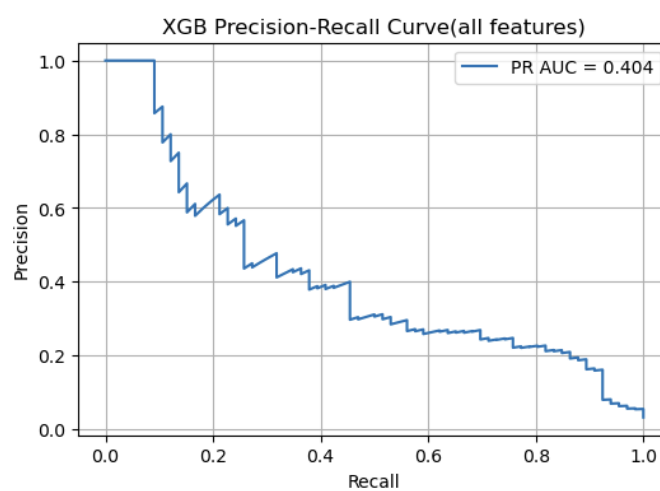


Figure 26: XGBoost PR-AUC Curve (all features)

XGB Distribution of Cross-Validation recall Scores(selected features)

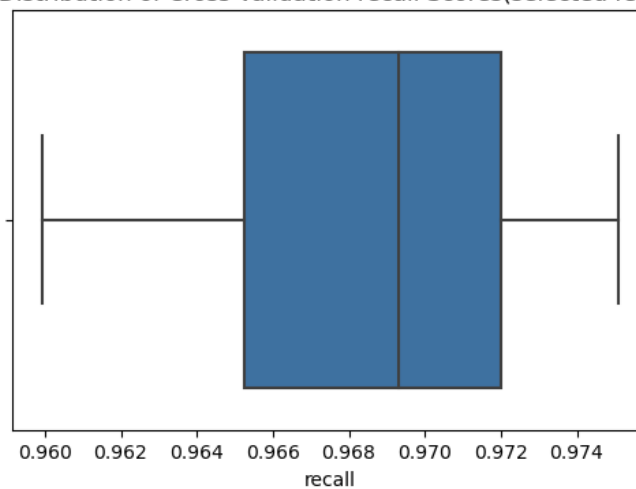


Figure 27: XGBoost Distribution of Cross-Validation recall scores (selected features)

XGB Distribution of Cross-Validation recall Scores(all features)

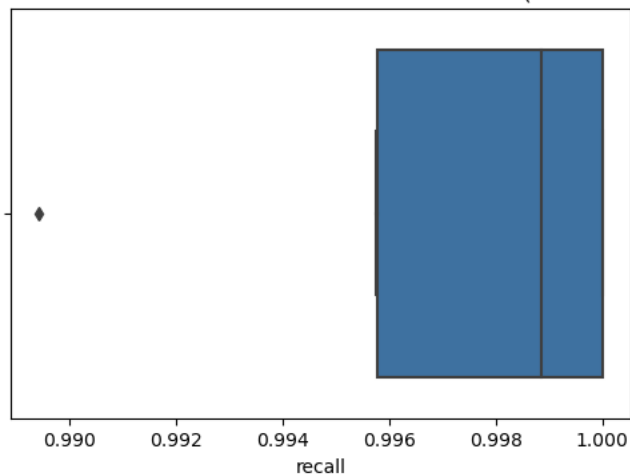


Figure 28: XGBoost Distribution of Cross-Validation recall scores (all features)

XGB Histogram of k-Fold recall Scores(selected features)

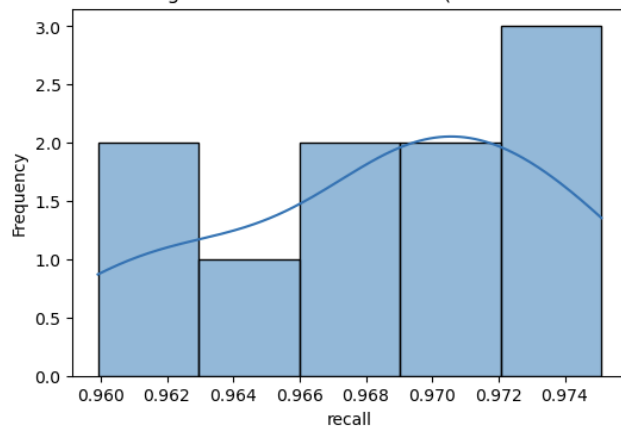


Figure 29: XGBoost Histogram of Cross-Validation recall scores (selected features)

XGB Histogram of k-Fold recall Scores(all features)

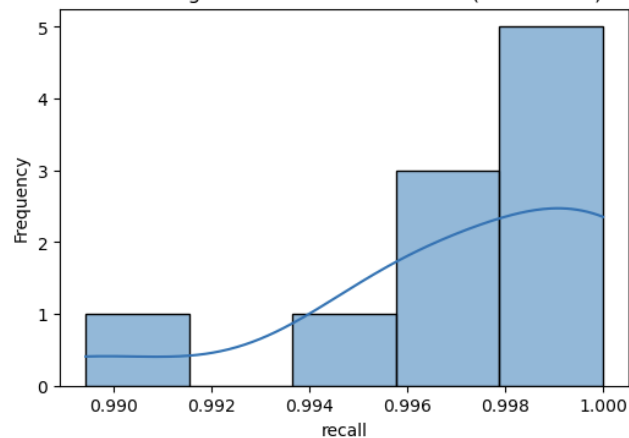


Figure 30: XGBoost Histogram of Cross-Validation recall scores (all features)

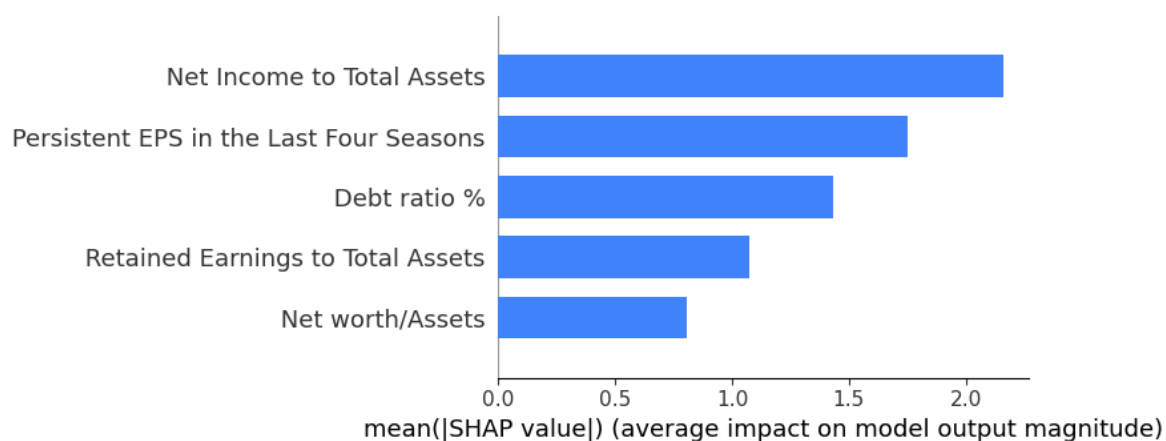


Figure 31: XGBoost SHAP Summary Plot (selected features)

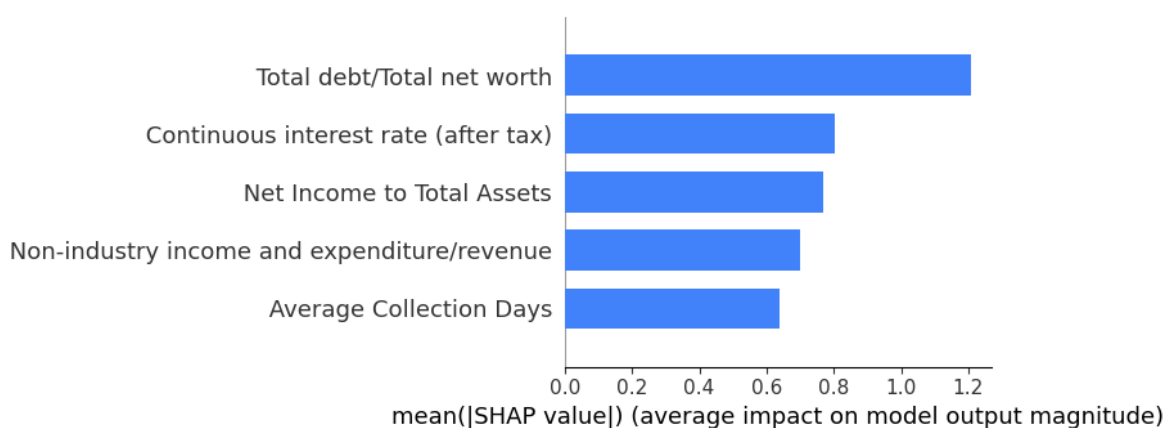


Figure 32: XGBoost SHAP Summary Plot (all features)

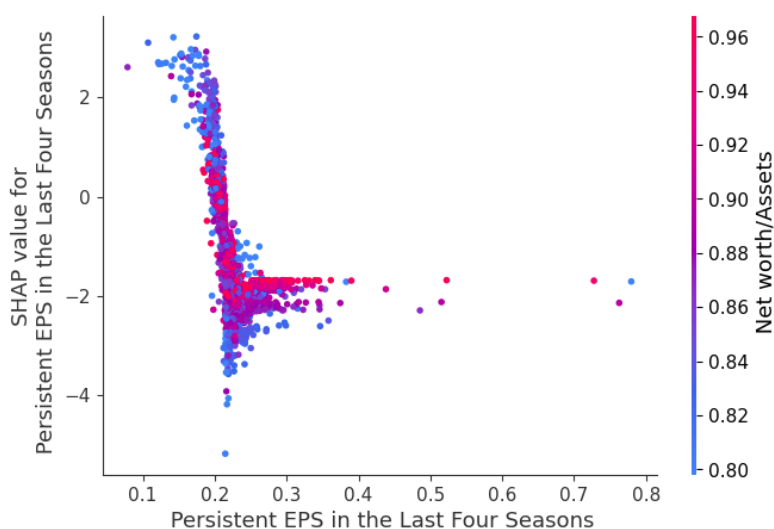


Figure 33: XGBoost SHAP Dependence Plot (selected features)

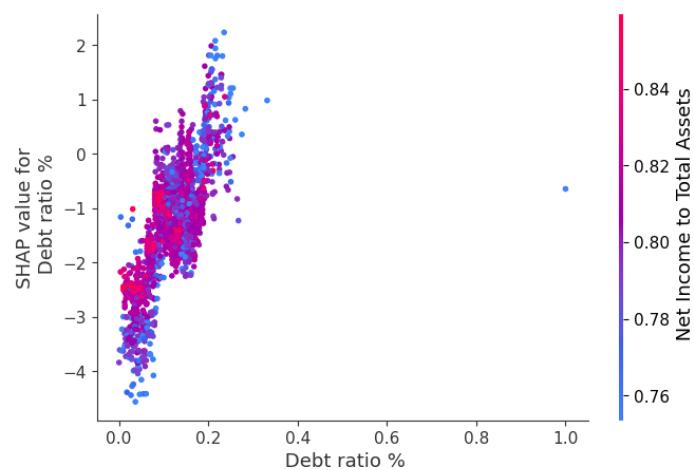


Figure 34: XGBoost SHAP Dependence Plot (selected features)

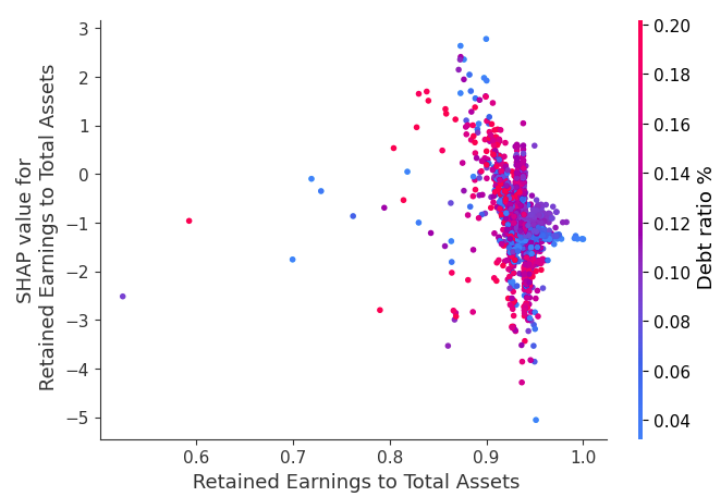


Figure 35: XGBoost SHAP Dependence Plot (selected features)

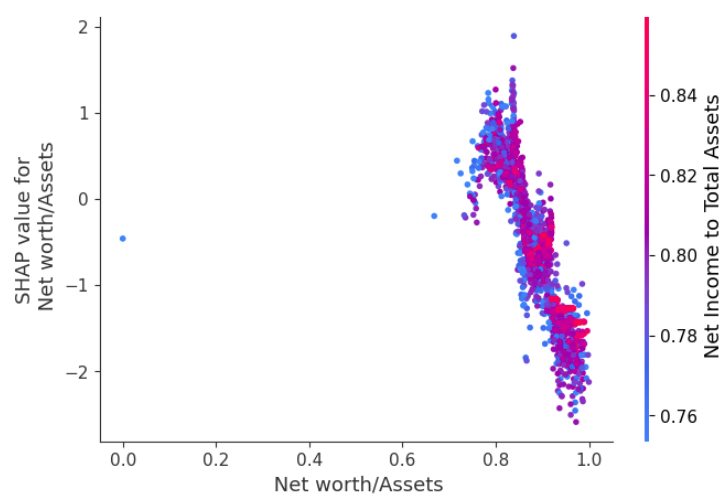


Figure 36: XGBoost SHAP Dependence Plot (selected features)

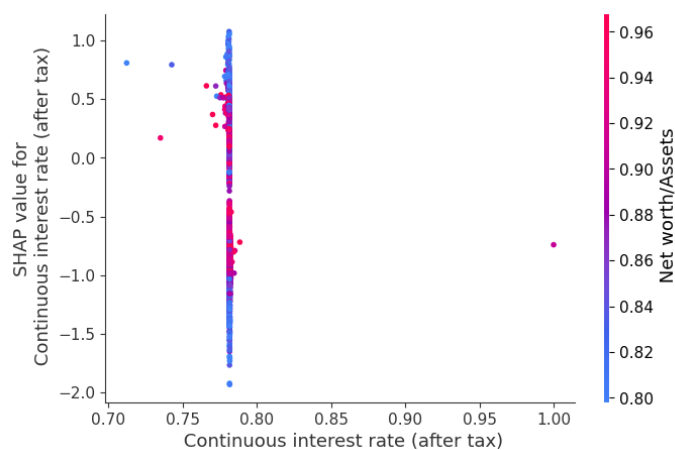


Figure 37: XGBoost SHAP Dependence Plot (all features)

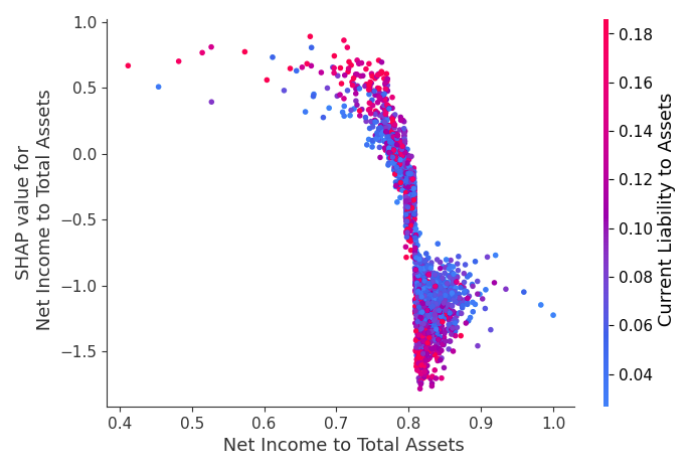


Figure 38: XGBoost SHAP Dependence Plot (all features)

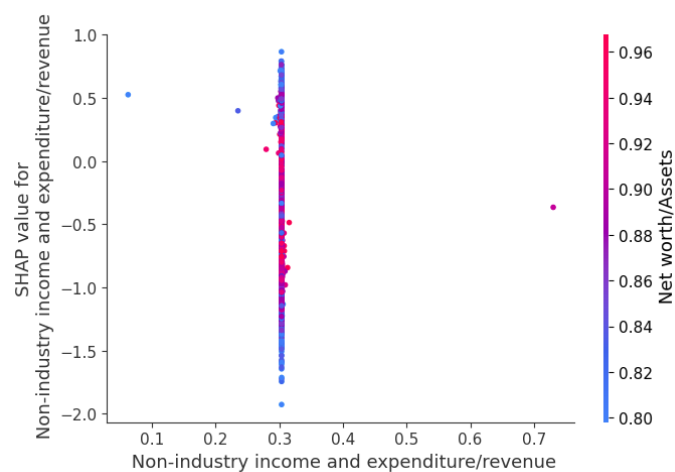


Figure 39: XGBoost SHAP Dependence Plot (all features)

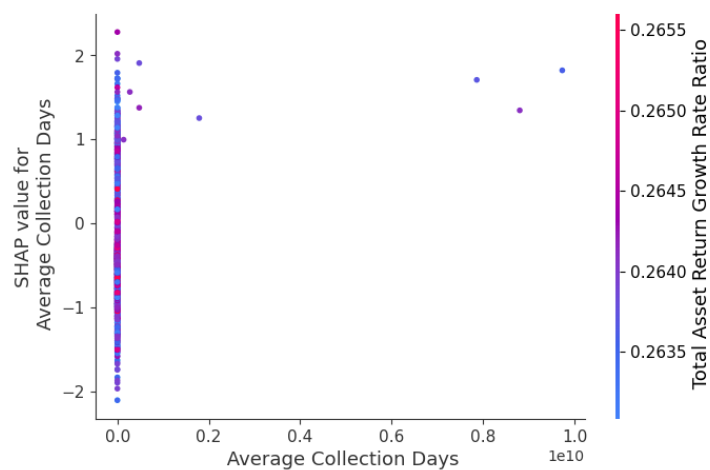


Figure 40: XGBoost SHAP Dependence Plot (all features)

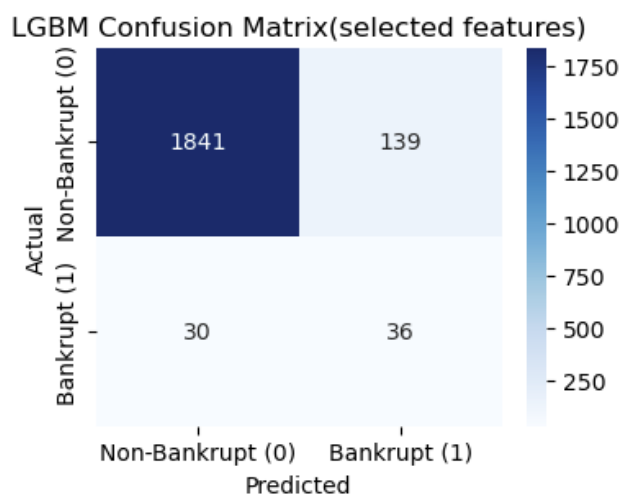


Figure 41: LightGBM Confusion Matrix (selected features)

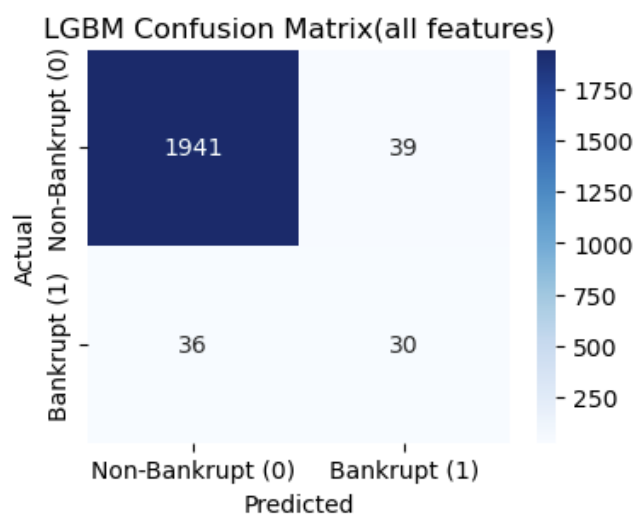


Figure 42: LightGBM Confusion Matrix (all features)

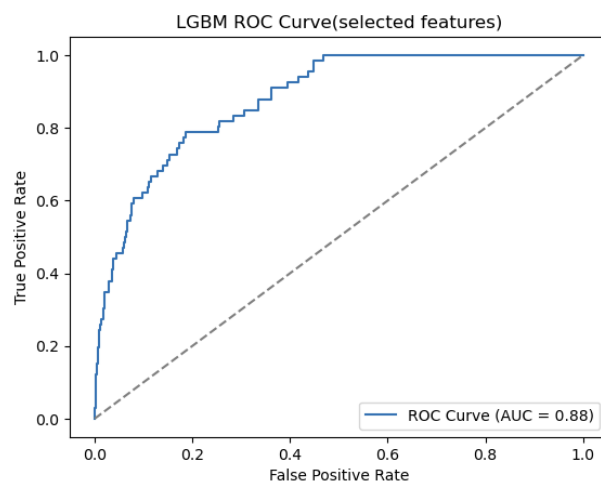


Figure 43: LightGBM ROC-AUC Curve (selected features)

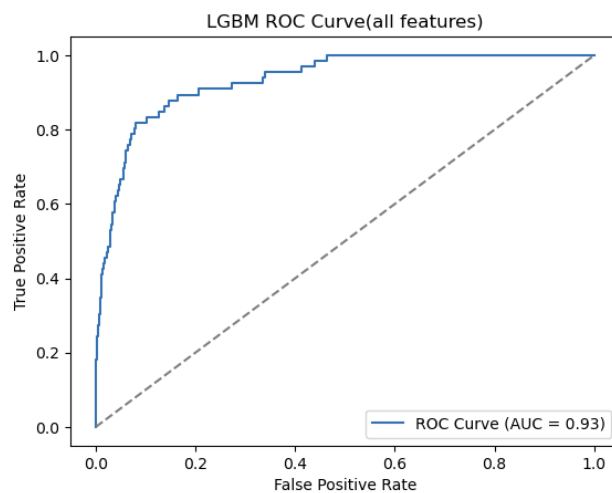


Figure 44: LightGBM ROC-AUC Curve (all features)

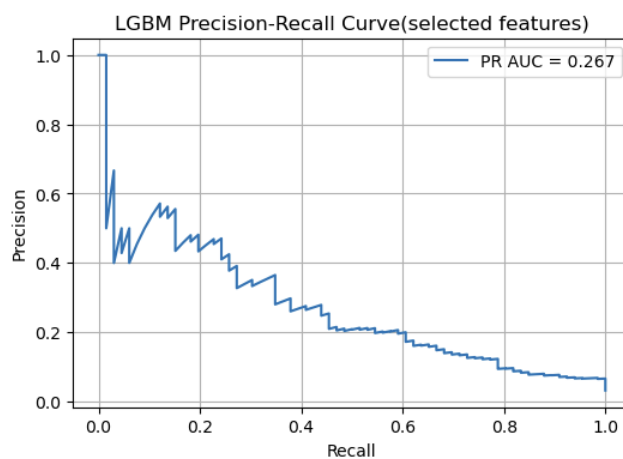


Figure 45: LightGBM PR-AUC Curve (selected features)

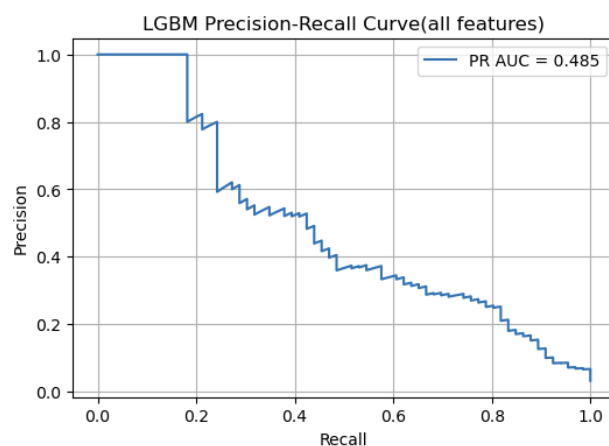


Figure 46: LightGBM PR-AUC Curve (all features)

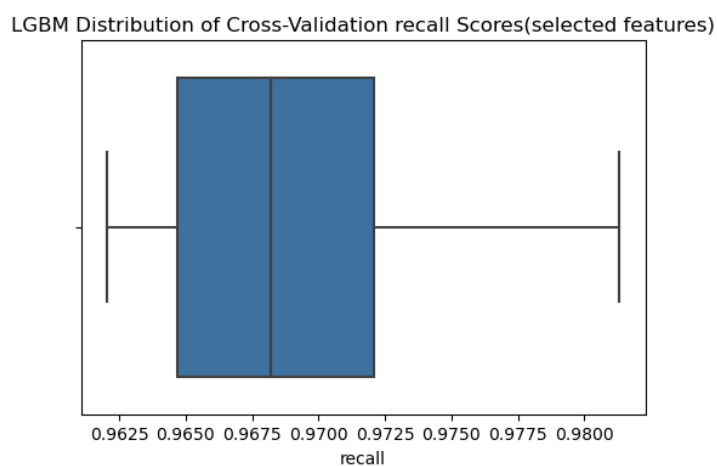


Figure 47: LightGBM Distribution of Cross-Validation recall scores (selected features)

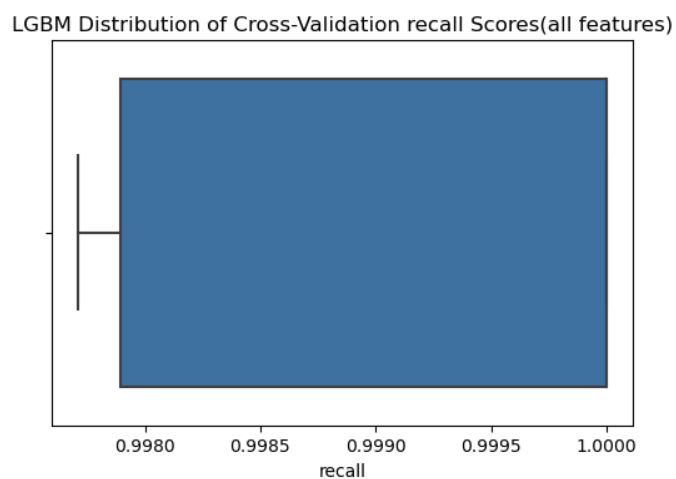


Figure 48: LightGBM Distribution of Cross-Validation recall scores (all features)

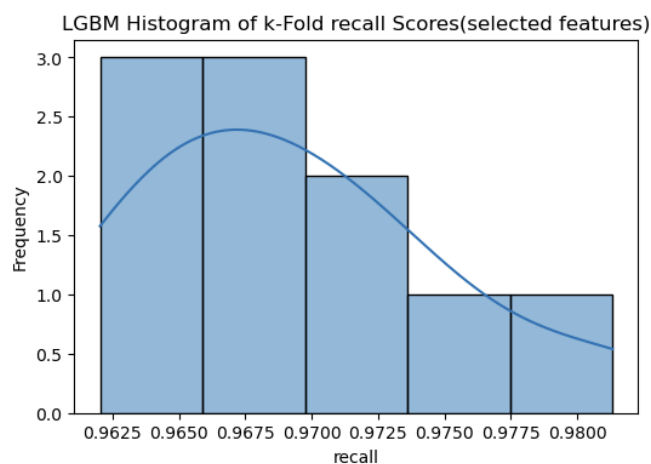


Figure 49: LightGBM Histogram of Cross-Validation recall scores (selected features)

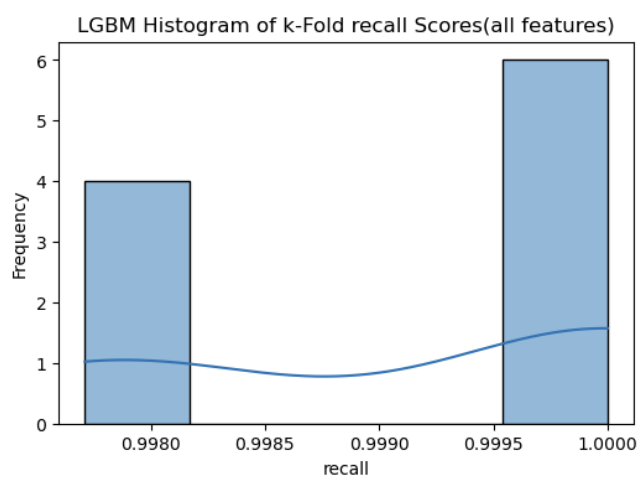


Figure 50: LightGBM Histogram of Cross-Validation recall scores (all features)

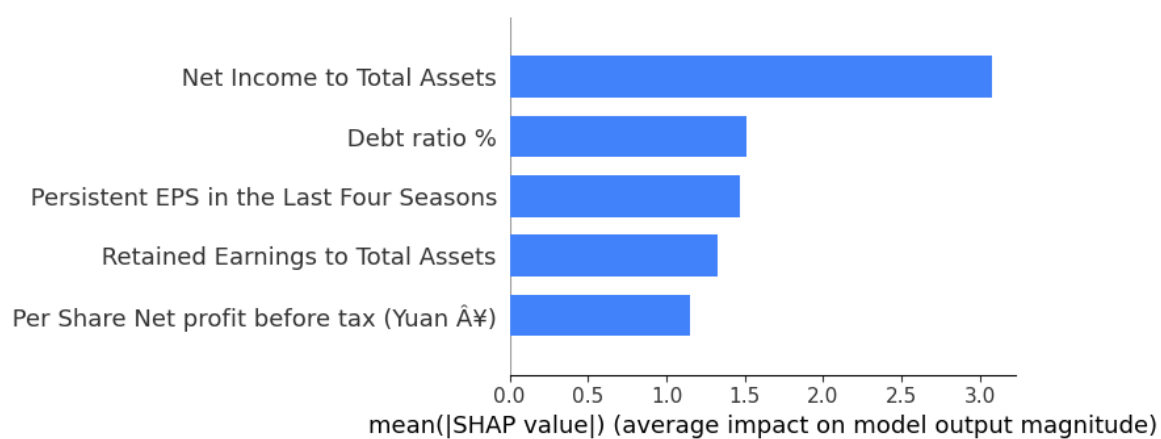


Figure 51: LightGBM SHAP Summary Plot (selected features)

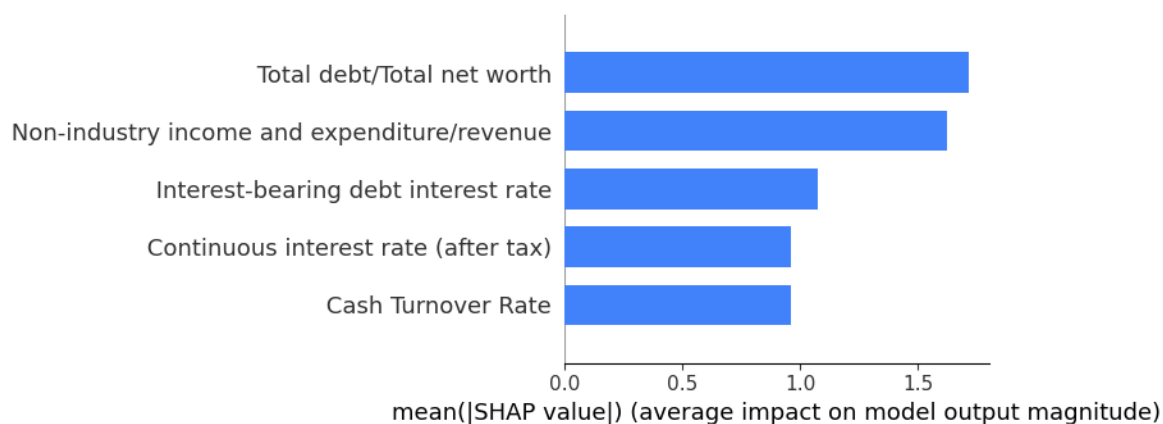


Figure 52: LightGBM SHAP Summary Plot (all features)

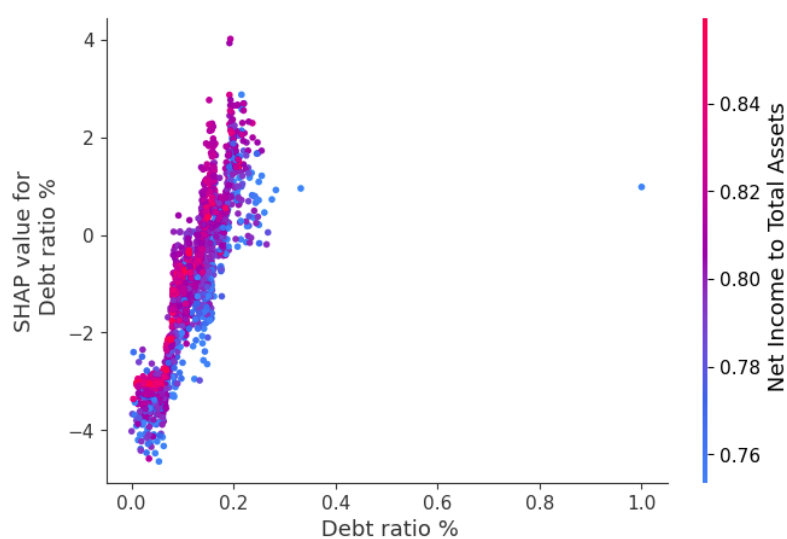


Figure 53: LightGBM SHAP Dependence Plot (selected features)

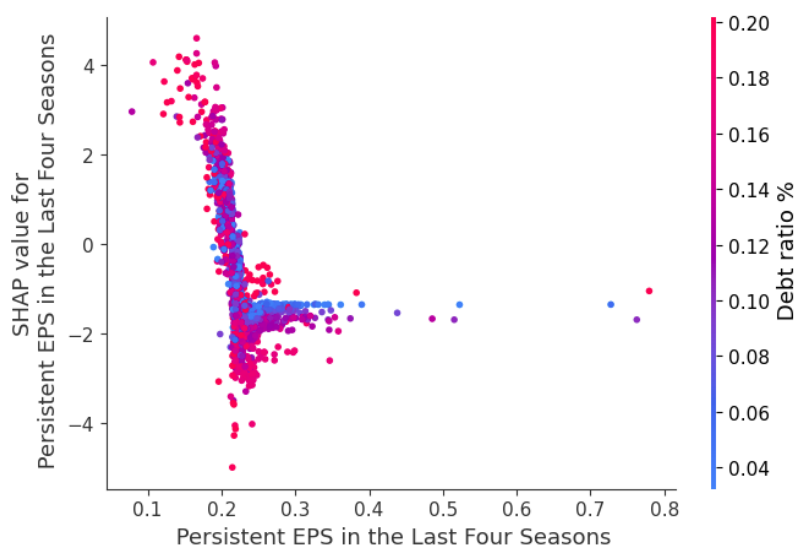


Figure 54: LightGBM SHAP Dependence Plot (selected features)

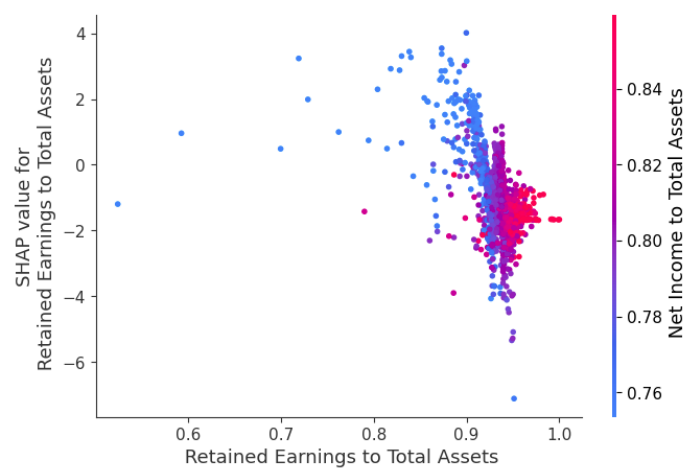


Figure 55: LightGBM SHAP Dependence Plot (selected features)

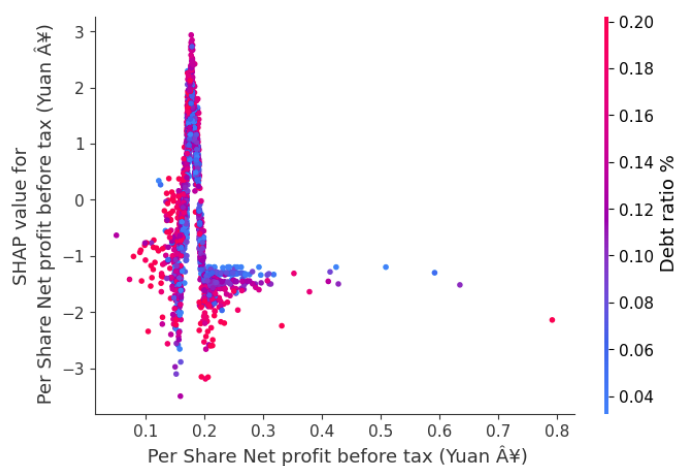


Figure 56: LightGBM SHAP Dependence Plot (selected features)

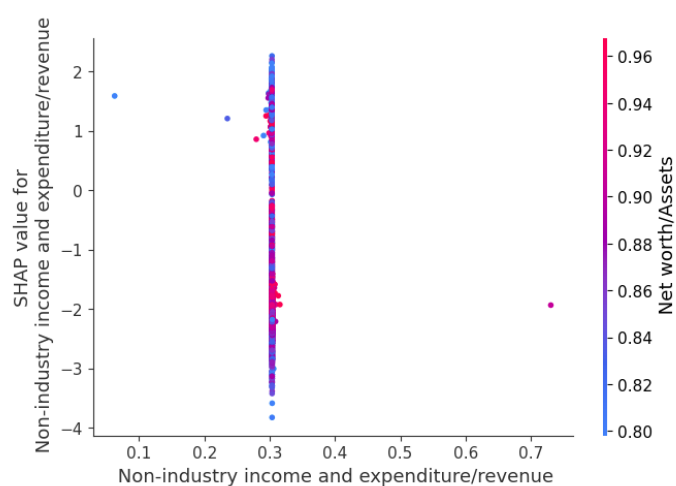


Figure 57: LightGBM SHAP Dependence Plot (all features)

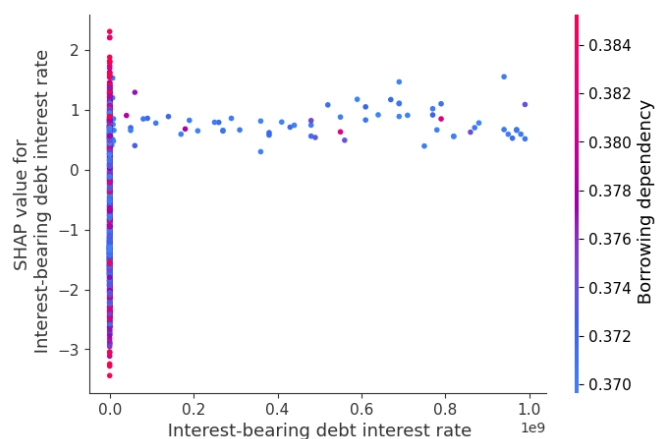


Figure 58: LightGBM SHAP Dependence Plot (all features)

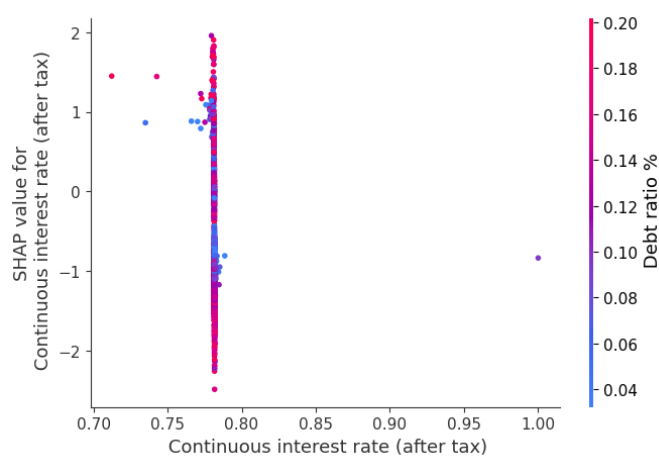


Figure 59: LightGBM SHAP Dependence Plot (all features)

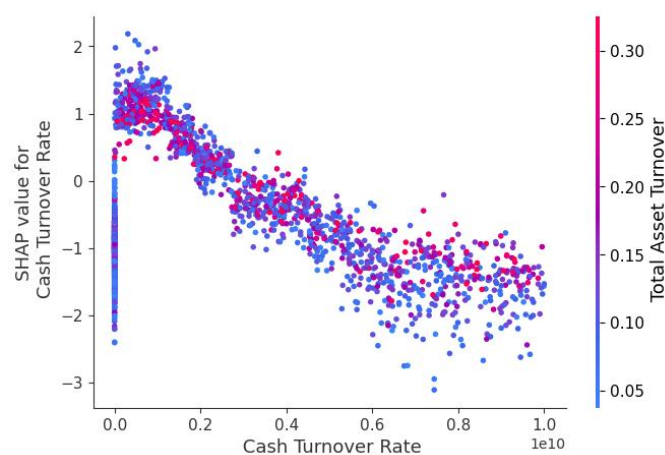


Figure 60: LightGBM SHAP Dependence Plot (all features)

DECLARATION OF AUTHENTICITY

I hereby declare that, the Master's thesis conducted independently and used only listed materials and sources are used. All materials used, from published as well as unpublished sources, whether directly quoted or paraphrased, are duly reported.

Furthermore, I declare that the Master's thesis, or any abridgment of it, was not used for any other degree seeking purpose.

(Signature)

.....
Md Rafi Bin Arman

Place, date